

STOCHASTIC DIFFERENTIAL GAMES AND VISCOSITY SOLUTIONS OF HAMILTON–JACOBI–BELLMAN–ISAACS EQUATIONS*

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Abstract. In this paper we study zero-sum two-player stochastic differential games with the help of the theory of backward stochastic differential equations (BSDEs). More precisely, we generalize the results of the pioneering work of Fleming and Souganidis [*Indiana Univ. Math. J.*, 38 (1989), pp. 293–314] by considering cost functionals defined by controlled BSDEs and by allowing the admissible control processes to depend on events occurring before the beginning of the game. This extension of the class of admissible control processes has the consequence that the cost functionals become random variables. However, by making use of a Girsanov transformation argument, which is new in this context, we prove that the upper and the lower value functions of the game remain deterministic. Apart from the fact that this extension of the class of admissible control processes is quite natural and reflects the behavior of the players who always use the maximum of available information, its combination with BSDE methods, in particular that of the notion of stochastic “backward semigroups” introduced by Peng [*BSDE and stochastic optimizations*, in *Topics in Stochastic Analysis*, Science Press, Beijing, 1997], allows us then to prove a dynamic programming principle for both the upper and the lower value functions of the game in a straightforward way. The upper and the lower value functions are then shown to be the unique viscosity solutions of the upper and the lower Hamilton–Jacobi–Bellman–Isaacs equations, respectively. For this Peng’s BSDE method is extended from the framework of stochastic control theory into that of stochastic differential games.

Key words. stochastic differential games, value function, backward stochastic differential equations, dynamic programming principle, viscosity solution

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1. Introduction. With their pioneering paper of 1989 Fleming and Souganidis [14] were the first to study in a rigorous manner two-player zero-sum stochastic differential games and to prove that the lower and the upper value functions of such games satisfy the dynamic programming principle and that they are the unique viscosity solutions of the associated Bellman–Isaacs equations and coincide under the Isaacs condition. Their work has translated former results on differential games by Isaacs [23], Friedman [15], and, in particular, Evans and Souganidis [13] from the purely deterministic into the stochastic framework and has given an important impulse for the research in the theory of stochastic differential games. The paper of Fleming and Souganidis [14] has been the starting point for a lot of works using their approach and translating it into new contexts. So, for instance, Buckdahn, Cardaliaguet, and Rainer [6] adapted the methods of [14] in order to prove the existence of Nash equilibrium points for stochastic nonzero-sum differential games and to characterize them.

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Another direction for generalization was chosen by Bayraktar and Poor [4]: They studied stochastic differential games whose modulating process is a fractional Brownian motion, while Browne [5] considered stochastic dynamic investment games in continuous time, provided conditions under which a general payoff function has an achievable value, and gave, under these conditions, an explicit representation for the value and resulting equilibrium strategies. Concerning optimal stopping games the interested reader is referred to the work of Ekström and Peskir [11], of Karatzas and Sudderth [21], as well as of Karatzas and Zamfirescu [22]. Finally, the reader more interested in the subject of stochastic differential games is also referred to the references given in [14].

The present work also investigates two-player zero-sum stochastic differential games, but with two main differences to the setting chosen by Fleming and Souganidis [14] and the other papers mentioned above: On the one hand, we allow our admissible control processes to depend on the full past of the trajectories of the driving Brownian motion; this means, in particular, that they can also depend on information occurring before the beginning of the game (which has the consequence that the cost functionals become random variables); on the other hand, we consider a more general running cost functional, which implies that the cost functionals will be given by a backward stochastic differential equation (BSDE). Both of these extensions of the framework in [14] are crucial because they allow one to harmonize the setting for stochastic differential games with that for the stochastic control theory and to simplify considerably the approach in [14] by using BSDE methods.

BSDEs in their general nonlinear form were introduced by Pardoux and Peng [24] in 1990. They have been studied since then by a lot of authors and have found various applications, namely, in stochastic control, finance, and the second order PDE theory. BSDE methods, originally developed by Peng [26], [27] for the stochastic control theory, have been introduced in the theory of stochastic differential games by Hamadene and Lepeltier [16] and Hamadene, Lepeltier, and Peng [17] to study games with a dynamics whose diffusion coefficient is strictly elliptic and does not depend on the controls. In our present work there is not any such restriction on the diffusion coefficient and the application of BSDE methods; in particular, the notion of stochastic backward semigroups (Peng [26]) allows us to prove the dynamic programming principle for the upper and lower value functions of the game in a very straightforward way (i.e., in particular without making use of r -strategies (see Definition 1.7 in [14]) and π -admissible strategies (see Definition 2.2 in [14]) playing an essential role in [14]) and to derive from it with the help of Peng's method (see [26], [27]) the associated Bellman–Isaacs equations.

The dynamics of the stochastic differential game we investigate is given by the controlled stochastic differential equation

$$(1.1) \quad \begin{cases} dX_s^{t,x;u,v} &= b(s, X_s^{t,x;u,v}, u_s, v_s)ds + \sigma(s, X_s^{t,x;u,v}, u_s, v_s)dB_s, \\ X_t^{t,x;u,v} &= x(\in \mathbb{R}^n), \end{cases} \quad s \in [t, T],$$

where $T > 0$ is an arbitrarily fixed finite time horizon, $B = (B_s)_{s \in [0, T]}$ is a d -dimensional standard Brownian motion, and $u = (u_s)_{s \in [t, T]}$, $v = (v_s)_{s \in [t, T]}$ are progressively measurable with respect to the Brownian filtration and take their values in some compact metric spaces U and V , respectively (we will say that $u \in \mathcal{U}_{t, T}$, $v \in \mathcal{V}_{t, T}$). Precise assumptions on the coefficients $b : [0, T] \times \mathbb{R}^n \times U \times V \rightarrow \mathbb{R}^n$ and $\sigma : [0, T] \times \mathbb{R}^n \times U \times V \rightarrow \mathbb{R}^{n \times d}$ are given in the next section.

The cost functional (interpreted as a payoff for player I and as a cost for player

II) is introduced by a BSDE:

$$(1.2) \quad \begin{cases} -dY_s^{t,x;u,v} &= f(s, X_s^{t,x;u,v}, Y_s^{t,x;u,v}, Z_s^{t,x;u,v}, u_s, v_s)ds - Z_s^{t,x;u,v}dB_s, \\ Y_T^{t,x;u,v} &= \Phi(X_T^{t,x;u,v}), \end{cases} \quad s \in [t, T],$$

where the driver $f : [0, T] \times \mathbb{R}^n \times \mathbb{R} \times \mathbb{R}^d \times U \times V \rightarrow \mathbb{R}$ describes the running cost and $\Phi : \mathbb{R}^n \rightarrow \mathbb{R}$ the terminal cost. Under the assumptions on f and Φ that will be introduced in the next section, the above BSDE has a unique solution $(Y_s^{t,x;u,v}, Z_s^{t,x;u,v})_{s \in [t, T]}$, and the cost functional is given by

$$(1.3) \quad J(t, x; u, v) = Y_t^{t,x;u,v}.$$

As usual in the differential game theory, the players cannot restrict to play only control processes, and one player has to fix a strategy while the other player chooses the best answer to this strategy in the form of a control process. A strategy admissible for player I (resp., player II) is a nonanticipating mapping $\alpha : \mathcal{V}_{t,T} \rightarrow \mathcal{U}_{t,T}$ (resp., $\beta : \mathcal{U}_{t,T} \rightarrow \mathcal{V}_{t,T}$) which associates every admissible control of the other player with one of his own admissible controls (we write: $\alpha \in \mathcal{A}_{t,T}, \beta \in \mathcal{B}_{t,T}$; the precise definitions can be found in section 3). We define the lower value function of our stochastic differential game as follows:

$$(1.4) \quad W(t, x) := \operatorname{essinf}_{\beta \in \mathcal{B}_{t,T}} \operatorname{esssup}_{u \in \mathcal{U}_{t,T}} J(t, x; u, \beta(u)),$$

and the upper value function is given by

$$(1.5) \quad U(t, x) := \operatorname{esssup}_{\alpha \in \mathcal{A}_{t,T}} \operatorname{essinf}_{v \in \mathcal{V}_{t,T}} J(t, x; \alpha(v), v).$$

The objective of our paper is to investigate these lower and upper value functions. The main results of the paper state that W and U are deterministic (Proposition 3.3) continuous viscosity solutions of the Bellman–Isaacs equations (Theorem 4.2)

$$(1.6) \quad \begin{cases} \frac{\partial}{\partial t} W(t, x) + H^-(t, x, W, DW, D^2W) = 0, & (t, x) \in [0, T] \times \mathbb{R}^n, \\ W(T, x) = \Phi(x), & x \in \mathbb{R}^n, \end{cases}$$

and

$$(1.7) \quad \begin{cases} \frac{\partial}{\partial t} U(t, x) + H^+(t, x, U, DU, D^2U) = 0, & (t, x) \in [0, T] \times \mathbb{R}^n, \\ U(T, x) = \Phi(x), & x \in \mathbb{R}^n, \end{cases}$$

respectively, associated with the Hamiltonians

$$H^-(t, x, y, p, X) = \sup_{u \in U} \inf_{v \in V} H(t, x, y, p, X, u, v),$$

$$H^+(t, x, y, p, X) = \inf_{v \in V} \sup_{u \in U} H(t, x, y, p, X, u, v),$$

$(t, x, y, p, X) \in [0, T] \times \mathbb{R}^n \times \mathbb{R} \times \mathbb{R}^n \times S^n$ (recall that S^n denotes the set of all $n \times n$ symmetric matrices), where

$$(1.8) \quad \begin{aligned} H(t, x, y, p, X, u, v) &= 1/2 \cdot \operatorname{tr}(\sigma \sigma^T(t, x, u, v)X) \\ &+ p \cdot b(t, x, u, v) + f(t, x, y, p \cdot \sigma(t, x, u, v), u, v). \end{aligned}$$

Moreover, we prove the uniqueness (Theorem 5.3) in a class of continuous functions with a growth condition which was introduced by Barles, Buckdahn, and Pardoux [3] and is weaker than the polynomial growth assumption. Fleming and Souganidis [14] obtained the above equations for stochastic differential games with the running cost $f(t, x, y, z, u, v) = f(t, x, u, v)$. Let us also mention that Cheridito et al. [7] discussed Bellman–Isaacs equations with $f(t, x, y, z, u, v) = \alpha(t, x, u, v) + \beta(t, x, u, v)y$ as an example of nonlinear PDEs to which they gave a stochastic control interpretation in terms of so-called 2-BSDEs and auxiliary stochastic target problems.

Notice the fact that W and U , introduced as a combination of essential infimum and essential supremum over a class of random variables, are deterministic is far from being trivial. The method developed by Peng [26], [27] (see also Theorem A.2 of the present paper) for value functions involving only control processes but not strategies does not apply here since the strategies from $\mathcal{A}_{t,T}$ and $\mathcal{B}_{t,T}$ do not have, in general, any continuity property. To overcome this difficulty we show in Proposition 3.3 and Lemma 3.4 that W and U are invariant under Girsanov transformation and use the fact that a functional of the Brownian motion which is invariant under Girsanov transformation into all directions of the Cameron–Martin space must be deterministic. We emphasize that the proof of Lemma 3.4 does not use BSDE methods which makes this method also applicable to other situations, such as standard stochastic control problems.

Our paper is organized as follows. Section 2 and Appendix A recall some elements of the theory of backward SDEs and forward-backward SDEs which will be needed in what follows. Section 3 introduces the setting of the stochastic differential game and, in particular, the notion of the lower and the upper value functions (see (1.4) and (1.5)). Both functions W and U , a priori random fields, turn out to be deterministic functions (Proposition 3.3) which are Lipschitz in x (Lemma 3.5) and $\frac{1}{2}$ -Hölder continuous in t (Theorem 3.10). Moreover, they satisfy the dynamic programming principle (DPP) (Theorem 3.6). The DPP allows us to derive in section 4 with the help of Peng’s method that W and U are viscosity solutions of the associated Bellman–Isaacs equations (Theorem 4.2). The uniqueness is studied in section 5. The main result of this section is a comparison result which compares viscosity sub- and supersolutions of Bellman–Isaacs equations (Theorem 5.3). This theorem not only allows us to characterize W and U as unique viscosity solutions of associated Bellman–Isaacs equations but also to show that, under the Isaacs condition, W and U coincide (one says that the game has a value; see Remark 5.2), while, in general, without the Isaacs condition we have only $W \leq U$ (see Remark 5.1). Finally, under the more restrictive assumptions of Fleming and Souganidis, we identify W and U with the value functions defined by them in [14]; see Remark 5.3.

2. Preliminaries. Let us first introduce the setting in which we want to study stochastic differential games. The probability space on which we work is the classical Wiener space $(\Omega, \mathcal{F}, P)$, and the driving Brownian motion B will be the coordinate process on Ω . Let us be more precise: Ω is the set of continuous functions from $[0, T]$ to $\mathbb{R}^d$ starting from 0 ($\Omega = C_0([0, T]; \mathbb{R}^d)$), $\mathcal{F}$ is the Borel σ -algebra over Ω , completed with respect to the Wiener measure P on this space, and B denotes the coordinate process: $B_s(\omega) = \omega_s$, $s \in [0, T]$, $\omega \in \Omega$. By $\mathbb{F} = \{\mathcal{F}_s, 0 \leq s \leq T\}$ we denote the natural filtration generated by $\{B_s\}_{0 \leq s \leq T}$ and augmented by all P -null sets, i.e.,

$$\mathcal{F}_s = \sigma\{B_r, r \leq s\} \vee \mathcal{N}_P, \quad s \in [0, T],$$

where $\mathcal{N}_P$ is the set of all P -null subsets and $T > 0$ a fixed real time horizon. For any $n \geq 1$, $|z|$ denotes the Euclidean norm of $z \in \mathbb{R}^n$. We also shall introduce the

following spaces of processes which will be used frequently in what follows:

$\mathcal{S}^2(0, T; \mathbb{R}) := \{(\psi_t)_{0 \leq t \leq T} \text{ real-valued adapted càdlàg process} :$

$$E[\sup_{0 \leq t \leq T} |\psi_t|^2] < +\infty\};$$

$\mathcal{H}^2(0, T; \mathbb{R}^n) := \{(\psi_t)_{0 \leq t \leq T} \mathbb{R}^n\text{-valued progressively measurable process} :$

$$\|\psi\|_2^2 = E[\int_0^T |\psi_t|^2 dt] < +\infty\}.$$

Let us now consider a function $g : \Omega \times [0, T] \times \mathbb{R} \times \mathbb{R}^d \rightarrow \mathbb{R}$ with the property that $(g(t, y, z))_{t \in [0, T]}$ is progressively measurable for each (y, z) in $\mathbb{R} \times \mathbb{R}^d$, and we also make the following assumptions on g throughout the paper:

(A1) There exists a constant $C \geq 0$ such that, P -a.s., for all $t \in [0, T]$, $y_1, y_2 \in \mathbb{R}$, $z_1, z_2 \in \mathbb{R}^d$,

$$|g(t, y_1, z_1) - g(t, y_2, z_2)| \leq C(|y_1 - y_2| + |z_1 - z_2|).$$

(A2) $g(\cdot, 0, 0) \in \mathcal{H}^2(0, T; \mathbb{R})$.

The following result on BSDEs is by now well known; for its proof the reader is referred to Pardoux and Peng [24].

LEMMA 2.1. *Under the assumptions (A1) and (A2), for any random variable $\xi \in L^2(\Omega, \mathcal{F}_T, P)$, the BSDE*

$$(2.1) \quad y_t = \xi + \int_t^T g(s, y_s, z_s) ds - \int_t^T z_s dB_s, \quad 0 \leq t \leq T,$$

has a unique adapted solution

$$(y_t^{T, g, \xi}, z_t^{T, g, \xi})_{t \in [0, T]} \in \mathcal{S}^2(0, T; \mathbb{R}) \times \mathcal{H}^2(0, T; \mathbb{R}^d).$$

In what follows, we always assume that the driving coefficient g of a BSDE satisfies (A1) and (A2).

Let us remark that Lemma 2.1 remains true when assumption (A1) is replaced by weaker assumptions, for instance those studied in Bahlali [1], Bahlali et al. [2], or Pardoux and Peng [25]. However, here, for the sake of simplicity of the calculus we prefer to work with the Lipschitz assumption.

We also shall recall the following basic results on BSDEs. We begin with the well-known comparison theorem (see Theorem 2.2 in El Karoui, Peng, and Quenez [12]).

LEMMA 2.2 (comparison theorem). *Given two coefficients g_1 and g_2 satisfying (A1) and (A2) and two terminal values $\xi_1, \xi_2 \in L^2(\Omega, \mathcal{F}_T, P)$, we denote by (y^1, z^1) and (y^2, z^2) the solution of BSDE with the data (ξ_1, g_1) and (ξ_2, g_2) , respectively. Then we have:*

(i) (Monotonicity). *If $\xi_1 \geq \xi_2$ and $g_1 \geq g_2$, a.s., then $y_t^1 \geq y_t^2$, a.s., for all $t \in [0, T]$.*

(ii) (Strict monotonicity). *If, in addition to (i), we also assume that $P(\xi_1 > \xi_2) > 0$, then $P\{y_t^1 > y_t^2\} > 0$, $0 \leq t \leq T$, and, in particular, $y_0^1 > y_0^2$.*

Using the notation introduced in Lemma 2.2 we now suppose that, for some $g : \Omega \times [0, T] \times \mathbb{R} \times \mathbb{R}^d \rightarrow \mathbb{R}$ satisfying (A1) and (A2) and for some $i \in \{1, 2\}$, the drivers g_i , $i = 1, 2$, are of the form

$$g_i(s, y_s^i, z_s^i) = g(s, y_s^i, z_s^i) + \varphi_i(s), \quad \text{d}P\text{-a.e., } i = 1, 2,$$

where $\varphi_i \in \mathcal{H}^2(0, T; \mathbb{R})$, $i = 1, 2$. Then, for terminal values ξ_1, ξ_2 belonging to $L^2(\Omega, \mathcal{F}_T, P)$ we have the following.

LEMMA 2.3. *The difference of the solutions (y^1, z^1) and (y^2, z^2) of BSDE with the data (ξ_1, g_1) and (ξ_2, g_2) , respectively, satisfies the following estimate:*

$$\begin{aligned} & |y_t^1 - y_t^2|^2 + \frac{1}{2} E \left[\int_t^T e^{\beta(s-t)} [|y_s^1 - y_s^2|^2 + |z_s^1 - z_s^2|^2] ds | \mathcal{F}_t \right] \\ & \leq E[e^{\beta(T-t)} |\xi_1 - \xi_2|^2 | \mathcal{F}_t] + E \left[\int_t^T e^{\beta(s-t)} |\varphi_1(s) - \varphi_2(s)|^2 ds | \mathcal{F}_t \right], \\ & \quad P\text{-a.s., for all } 0 \leq t \leq T, \end{aligned}$$

where $\beta = 16(1 + C^2)$.

Proof. For the proof the reader is referred to Proposition 2.1 in El Karoui, Peng, and Quenez [12] or Theorem 2.3 in Peng [26]. $\square$

3. Stochastic differential games and associated dynamic programming principles. Now we can study our stochastic differential games. The set of admissible control processes $\mathcal{U}$ (resp., $\mathcal{V}$) for the first (resp., second) player is the set of all U (resp., V)-valued $\mathcal{F}_t$ -progressively measurable processes. The control state spaces U and V are supposed to be compact metric spaces.

For given admissible controls $u(\cdot) \in \mathcal{U}$ and $v(\cdot) \in \mathcal{V}$, the according orbit which regards t as the initial time and $\zeta \in L^2(\Omega, \mathcal{F}_t, P; \mathbb{R}^n)$ as the initial state is defined by the solution of the following SDE:

$$(3.1) \quad \begin{cases} dX_s^{t, \zeta; u, v} &= b(s, X_s^{t, \zeta; u, v}, u_s, v_s) ds + \sigma(s, X_s^{t, \zeta; u, v}, u_s, v_s) dB_s, \quad s \in [t, T], \\ X_t^{t, \zeta; u, v} &= \zeta, \end{cases}$$

where the mappings

$$b : [0, T] \times \mathbb{R}^n \times U \times V \rightarrow \mathbb{R}^n \text{ and } \sigma : [0, T] \times \mathbb{R}^n \times U \times V \rightarrow \mathbb{R}^{n \times d}$$

satisfy the following conditions:

(H3.1)

- (i) For every fixed $x \in \mathbb{R}^n$, $b(\cdot, x, \cdot, \cdot)$ and $\sigma(\cdot, x, \cdot, \cdot)$ are continuous in (t, u, v) ;
- (ii) there exists a $C > 0$ such that, for all $t \in [0, T]$, $x, x' \in \mathbb{R}^n$, $u \in U$, $v \in V$, $|b(t, x, u, v) - b(t, x', u, v)| + |\sigma(t, x, u, v) - \sigma(t, x', u, v)| \leq C|x - x'|$.

From (H3.1) we can get the global linear growth conditions of b and σ , i.e., the existence of some $C > 0$ such that, for all $0 \leq t \leq T$, $u \in U$, $v \in V$, $x \in \mathbb{R}^n$,

$$(3.2) \quad |b(t, x, u, v)| + |\sigma(t, x, u, v)| \leq C(1 + |x|).$$

It follows from (A.2) in the Appendix that, under the above assumptions, for any $u(\cdot) \in \mathcal{U}$ and $v(\cdot) \in \mathcal{V}$, SDE (3.1) has a unique strong solution. Moreover, for any $p \geq 2$, there exists $C_p \in \mathbb{R}$ such that, for any $t \in [0, T]$, $u(\cdot) \in \mathcal{U}$, $v(\cdot) \in \mathcal{V}$, and $\zeta, \zeta' \in L^p(\Omega, \mathcal{F}_t, P; \mathbb{R}^n)$, we also have the following estimates, P -a.s.:

$$(3.3) \quad \begin{aligned} E \left[\sup_{s \in [t, T]} |X_s^{t, \zeta; u, v} - X_s^{t, \zeta'; u, v}|^p | \mathcal{F}_t \right] &\leq C_p |\zeta - \zeta'|^p, \\ E \left[\sup_{s \in [t, T]} |X_s^{t, \zeta; u, v}|^p | \mathcal{F}_t \right] &\leq C_p (1 + |\zeta|^p). \end{aligned}$$

The constant C_p depends only on the Lipschitz and the linear growth constants of b and σ with respect to x .

Let now be given two functions

$$\Phi : \mathbb{R}^n \rightarrow \mathbb{R}, \quad f : [0, T] \times \mathbb{R}^n \times \mathbb{R} \times \mathbb{R}^d \times U \times V \rightarrow \mathbb{R}$$

that satisfy the following conditions:

(H3.2)

- (i) For every fixed $(x, y, z) \in \mathbb{R}^n \times \mathbb{R} \times \mathbb{R}^d$, $f(\cdot, x, y, z, \cdot, \cdot)$ is continuous in (t, u, v) , and there exists a constant $C > 0$ such that, for all $t \in [0, T]$, $x, x' \in \mathbb{R}^n$, $y, y' \in \mathbb{R}$, $z, z' \in \mathbb{R}^d$, $u \in U$, and $v \in V$,

$$|f(t, x, y, z, u, v) - f(t, x', y', z', u, v)| \leq C(|x - x'| + |y - y'| + |z - z'|);$$

- (ii) there is a constant $C > 0$ such that, for all $x, x' \in \mathbb{R}^n$,

$$|\Phi(x) - \Phi(x')| \leq C|x - x'|.$$

From (H3.2) we see that f and Φ also satisfy the global linear growth condition in x , i.e., there exists some $C > 0$ such that, for all $0 \leq t \leq T$, $u \in U$, $v \in V$, and $x \in \mathbb{R}^n$,

$$(3.4) \quad |f(t, x, 0, 0, u, v)| + |\Phi(x)| \leq C(1 + |x|).$$

For any $u(\cdot) \in \mathcal{U}$, $v(\cdot) \in \mathcal{V}$, and $\zeta \in L^2(\Omega, \mathcal{F}_t, P; \mathbb{R}^n)$, the mappings $\xi := \Phi(X_T^{t, \zeta; u, v})$ and $g(s, y, z) := f(s, X_s^{t, \zeta; u, v}, y, z, u_s, v_s)$ satisfy the conditions of Lemma 2.1 on the interval $[t, T]$. Therefore, there exists a unique solution to the following BSDE:

$$(3.5) \quad \begin{cases} -dY_s^{t, \zeta; u, v} &= f(s, X_s^{t, \zeta; u, v}, Y_s^{t, \zeta; u, v}, Z_s^{t, \zeta; u, v}, u_s, v_s)ds - Z_s^{t, \zeta; u, v}dB_s, \\ Y_T^{t, \zeta; u, v} &= \Phi(X_T^{t, \zeta; u, v}), \end{cases}$$

where $X^{t, \zeta; u, v}$ is introduced by (3.1).

Moreover, in analogy to Proposition A.1, we can see that there exists some constant $C > 0$ such that, for all $0 \leq t \leq T$, $\zeta, \zeta' \in L^2(\Omega, \mathcal{F}_t, P; \mathbb{R}^n)$, $u(\cdot) \in \mathcal{U}$, and $v(\cdot) \in \mathcal{V}$, P -a.s.,

$$(3.6) \quad \begin{aligned} \text{(i)} \quad & |Y_t^{t, \zeta; u, v} - Y_t^{t, \zeta'; u, v}| \leq C|\zeta - \zeta'|; \\ \text{(ii)} \quad & |Y_t^{t, \zeta; u, v}| \leq C(1 + |\zeta|). \end{aligned}$$

We now introduce the following subspaces of admissible controls.

DEFINITION 3.1. An admissible control process $u = \{u_r, r \in [t, s]\}$ (resp., $v = \{v_r, r \in [t, s]\}$) for player I (resp., II) on $[t, s]$ ($t < s \leq T$) is an $\mathcal{F}_r$ -progressively measurable process taking values in U (resp., V). The set of all admissible controls for player I (resp., II) on $[t, s]$ is denoted by $\mathcal{U}_{t, s}$ (resp., $\mathcal{V}_{t, s}$). We identify two processes u and $\bar{u}$ in $\mathcal{U}_{t, s}$ and write $u \equiv \bar{u}$ on $[t, s]$, if $P\{u = \bar{u} \text{ a.e. in } [t, s]\} = 1$. Similarly we interpret $v \equiv \bar{v}$ on $[t, s]$ in $\mathcal{V}_{t, s}$.

Finally, we have still to define the admissible strategies for the game.

DEFINITION 3.2. A nonanticipative strategy for player I on $[t, s]$ ($t < s \leq T$) is a mapping $\alpha : \mathcal{V}_{t, s} \rightarrow \mathcal{U}_{t, s}$ such that, for any $\mathbb{F}$ -stopping time $S : \Omega \rightarrow [t, s]$ and any $v_1, v_2 \in \mathcal{V}_{t, s}$, with $v_1 \equiv v_2$ on $[t, S]$, it holds that $\alpha(v_1) \equiv \alpha(v_2)$ on $[t, S]$. Nonanticipative strategies for player II on $[t, s]$, $\beta : \mathcal{U}_{t, s} \rightarrow \mathcal{V}_{t, s}$, are defined similarly. The set of all nonanticipative strategies $\alpha : \mathcal{V}_{t, s} \rightarrow \mathcal{U}_{t, s}$ for player I on $[t, s]$ is denoted by $\mathcal{A}_{t, s}$. The set of all nonanticipative strategies $\beta : \mathcal{U}_{t, s} \rightarrow \mathcal{V}_{t, s}$ for player II on $[t, s]$ is denoted by $\mathcal{B}_{t, s}$. (Recall that $[t, S] = \{(r, \omega) \in [0, T] \times \Omega, t \leq r \leq S(\omega)\}$.)

Given the control processes $u(\cdot) \in \mathcal{U}_{t,T}$ and $v(\cdot) \in \mathcal{V}_{t,T}$ we introduce the following associated cost functional:

$$(3.7) \quad J(t, x; u, v) := Y_t^{t, x; u, v}, \quad (t, x) \in [0, T] \times \mathbb{R}^n,$$

where the process $Y^{t, x; u, v}$ is defined by BSDE (3.5).

Similarly to the proof of Theorem A.2 we can get that, for any $t \in [0, T]$ and $\zeta \in L^2(\Omega, \mathcal{F}_t, P; \mathbb{R}^n)$,

$$(3.8) \quad J(t, \zeta; u, v) = Y_t^{t, \zeta; u, v}, \quad P\text{-a.s.}$$

Being particularly interested in the case of a deterministic ζ , i.e., $\zeta = x \in \mathbb{R}^n$, we define the lower value function of our stochastic differential game

$$(3.9) \quad W(t, x) := \operatorname{ess\,inf}_{\beta \in \mathcal{B}_{t,T}} \operatorname{ess\,sup}_{u \in \mathcal{U}_{t,T}} J(t, x; u, \beta(u))$$

and its upper value function

$$(3.10) \quad U(t, x) := \operatorname{ess\,sup}_{\alpha \in \mathcal{A}_{t,T}} \operatorname{ess\,inf}_{v \in \mathcal{V}_{t,T}} J(t, x; \alpha(v), v).$$

The names “lower value function” and “upper value function” for W and U , respectively, are justified later by Remark 5.1.

Remark 3.1. (1) Here the essential infimum and the essential supremum should be understood as one with respect to indexed families of random variables (see, e.g., Dunford and Schwartz [10], Dellacherie [9], or the appendix in Karatzas and Shreve [20] for detailed discussions). For the convenience of the reader we recall the notion of $\operatorname{ess\,inf}$ of processes. Given a family of real-valued random variables η_α , $\alpha \in I$, a random variable η is said to be $\operatorname{ess\,inf}_{\alpha \in I} \eta_\alpha$, if

- (i) $\eta \leq \eta_\alpha$, P -a.s., for any $\alpha \in I$;
- (ii) if there is another random variable ξ such that $\xi \leq \eta_\alpha$, P -a.s., for any $\alpha \in I$, then $\xi \leq \eta$, P -a.s.

The random variable $\operatorname{ess\,sup}_{\alpha \in I} \eta_\alpha$ can be introduced now by the relation

$$\operatorname{ess\,sup}_{\alpha \in I} \eta_\alpha = -\operatorname{ess\,inf}_{\alpha \in I} (-\eta_\alpha).$$

Finally, recall that $\operatorname{ess\,inf}_{\alpha \in I} \eta_\alpha = \inf_{n \geq 1} \eta_{\alpha_n}$ for some countable family $(\alpha_n) \subset I$; $\operatorname{ess\,sup}_{\alpha \in I} \eta_\alpha$ has the same property.

(2) Obviously, under the assumptions (H3.1)–(H3.2), the lower value function $W(t, x)$ as well as the upper value function $U(t, x)$ are well-defined, and a priori they both are bounded $\mathcal{F}_t$ -measurable random variables. But it turns out that $W(t, x)$ and $U(t, x)$ are even deterministic. Indeed, concentrating on the study of the properties of $W(t, x)$ (the function $U(t, x)$ can be analyzed in the same manner), we can state the following,

PROPOSITION 3.3. *For any $(t, x) \in [0, T] \times \mathbb{R}^n$, we have $W(t, x) = E[W(t, x)]$, P -a.s. By identifying $W(t, x)$ with its deterministic version $E[W(t, x)]$ we can consider $W : [0, T] \times \mathbb{R}^n \rightarrow \mathbb{R}$ as a deterministic function.*

Remark 3.2. Recall that the fact that the lower and the upper value functions defined by Fleming and Souganidis [14] are deterministic is an immediate consequence of their definition. Indeed, for a game over the time interval $[t, T]$ only control processes which are independent of the past $\mathcal{F}_t$ are considered as admissible, and since the admissible strategies are supposed to associate admissible control processes of

one player with those of the other player, all of the associated cost functionals are independent of $\mathcal{F}_t$ and hence deterministic.

Proof. Let H denote the Cameron–Martin space of all absolutely continuous elements $h \in \Omega$ whose derivative $\dot{h}$ belongs to $L^2([0, T], \mathbb{R}^d)$.

For any $h \in H$, we define the mapping $\tau_h \omega := \omega + h$, $\omega \in \Omega$. Obviously, $\tau_h : \Omega \rightarrow \Omega$ is a bijection, and its law is given by $P \circ [\tau_h]^{-1} = \exp\{\int_0^T \dot{h}_s dB_s - \frac{1}{2} \int_0^T |\dot{h}_s|^2 ds\} P$. Let $(t, x) \in [0, T] \times \mathbb{R}^n$ be arbitrarily fixed, and put $H_t = \{h \in H | h(\cdot) = h(\cdot \wedge t)\}$. We split now the proof in the following steps:

First step: For any $u \in \mathcal{U}_{t,T}$, $v \in \mathcal{V}_{t,T}$, and $h \in H_t$, $J(t, x; u, v)(\tau_h) = J(t, x; u(\tau_h), v(\tau_h))$, P -a.s.

Indeed, we apply the Girsanov transformation to SDE (3.1) (with $\zeta = x$) and compare the obtained equation with the SDE obtained from (3.1) by substituting the transformed control processes $u(\tau_h)$ and $v(\tau_h)$ for u and v . Then from the uniqueness of the solution of (3.1) we get $X_s^{t,x;u,v}(\tau_h) = X_s^{t,x;u(\tau_h),v(\tau_h)}$ for any $s \in [t, T]$, P -a.s. Furthermore, by a similar Girsanov transformation argument we get from the uniqueness of the solution of BSDE (3.5)

$$Y_s^{t,x;u,v}(\tau_h) = Y_s^{t,x;u(\tau_h),v(\tau_h)} \text{ for any } s \in [t, T], \text{ } P\text{-a.s.},$$

$$Z_s^{t,x;u,v}(\tau_h) = Z_s^{t,x;u(\tau_h),v(\tau_h)}, \text{ d}sdP\text{-a.e. on } [t, T] \times \Omega.$$

That means

$$J(t, x; u, v)(\tau_h) = J(t, x; u(\tau_h), v(\tau_h)), \text{ } P\text{-a.s.}$$

Second step: For $\beta \in \mathcal{B}_{t,T}$, $h \in H_t$, let $\beta^h(u) := \beta(u(\tau_{-h}))(\tau_h)$, $u \in \mathcal{U}_{t,T}$. Then $\beta^h \in \mathcal{B}_{t,T}$.

Obviously, β^h maps $\mathcal{U}_{t,T}$ into $\mathcal{V}_{t,T}$. Moreover, this mapping is nonanticipating. Indeed, let $S : \Omega \rightarrow [t, T]$ be an $\mathbb{F}$ -stopping time and $u_1, u_2 \in \mathcal{U}_{t,T}$, with $u_1 \equiv u_2$ on $\llbracket t, S \rrbracket$. Then, obviously, $u_1(\tau_{-h}) \equiv u_2(\tau_{-h})$ on $\llbracket t, S(\tau_{-h}) \rrbracket$ (notice that $S(\tau_{-h})$ is still a stopping time), and because $\beta \in \mathcal{B}_{t,T}$ we have $\beta(u_1(\tau_{-h})) \equiv \beta(u_2(\tau_{-h}))$ on $\llbracket t, S(\tau_{-h}) \rrbracket$. Therefore,

$$\beta^h(u_1) = \beta(u_1(\tau_{-h}))(\tau_h) \equiv \beta(u_2(\tau_{-h}))(\tau_h) = \beta^h(u_2) \text{ on } \llbracket t, S \rrbracket.$$

Third step: For all $h \in H_t$ and $\beta \in \mathcal{B}_{t,T}$ we have

$$\{\text{esssup}_{u \in \mathcal{U}_{t,T}} J(t, x; u, \beta(u))\}(\tau_h) = \text{esssup}_{u \in \mathcal{U}_{t,T}} \{J(t, x; u, \beta(u))(\tau_h)\}, \text{ } P\text{-a.s.}$$

Indeed, with the notation $I(t, x, \beta) := \text{esssup}_{u \in \mathcal{U}_{t,T}} J(t, x; u, \beta(u))$, $\beta \in \mathcal{B}_{t,T}$, we have $I(t, x, \beta) \geq J(t, x; u, \beta(u))$, and thus $I(t, x, \beta)(\tau_h) \geq J(t, x; u, \beta(u))(\tau_h)$, P -a.s., for all $u \in \mathcal{U}_{t,T}$. On the other hand, for any random variable ζ satisfying $\zeta \geq J(t, x; u, \beta(u))(\tau_h)$, and hence also $\zeta(\tau_{-h}) \geq J(t, x; u, \beta(u))$, P -a.s., for all $u \in \mathcal{U}_{t,T}$, we have $\zeta(\tau_{-h}) \geq I(t, x, \beta)$, P -a.s., i.e., $\zeta \geq I(t, x, \beta)(\tau_h)$, P -a.s. Consequently,

$$I(t, x, \beta)(\tau_h) = \text{esssup}_{u \in \mathcal{U}_{t,T}} \{J(t, x; u, \beta(u))(\tau_h)\}, \text{ } P\text{-a.s.}$$

Fourth step: $W(t, x)$ is invariant with respect to the Girsanov transformation τ_h , i.e.,

$$W(t, x)(\tau_h) = W(t, x), \text{ } P\text{-a.s.}, \text{ for any } h \in H.$$

Indeed, similarly to the third step we can show that for all $h \in H_t$

$$\{\text{essinf}_{\beta \in \mathcal{B}_{t,T}} I(t, x; \beta)\}(\tau_h) = \text{essinf}_{\beta \in \mathcal{B}_{t,T}} \{I(t, x; \beta)(\tau_h)\}, \quad P\text{-a.s.}$$

Then, from the first step to the third step we have, for any $h \in H_t$,

$$\begin{aligned} W(t, x)(\tau_h) &= \text{essinf}_{\beta \in \mathcal{B}_{t,T}} \text{esssup}_{u \in \mathcal{U}_{t,T}} \{J(t, x; u, \beta(u))(\tau_h)\} \\ &= \text{essinf}_{\beta \in \mathcal{B}_{t,T}} \text{esssup}_{u \in \mathcal{U}_{t,T}} J(t, x; u(\tau_h), \beta^h(u(\tau_h))) \\ &= \text{essinf}_{\beta \in \mathcal{B}_{t,T}} \text{esssup}_{u \in \mathcal{U}_{t,T}} J(t, x; u, \beta^h(u)) \\ &= \text{essinf}_{\beta \in \mathcal{B}_{t,T}} \text{esssup}_{u \in \mathcal{U}_{t,T}} J(t, x; u, \beta(u)) \\ &= W(t, x), \quad P\text{-a.s.}, \end{aligned}$$

where we have used $\{u(\tau_h)|u(\cdot) \in \mathcal{U}_{t,T}\} = \mathcal{U}_{t,T}$ and $\{\beta^h|\beta \in \mathcal{B}_{t,T}\} = \mathcal{B}_{t,T}$ in order to obtain both latter equalities. Therefore, for any $h \in H_t$, $W(t, x)(\tau_h) = W(t, x)$, P -a.s., and since $W(t, x)$ is $\mathcal{F}_t$ -measurable, we have this relation even for all $h \in H$. Indeed, recall that our underlying fundamental space is $\Omega = C_0([0, T]; \mathbb{R}^d)$ and that, due to the definition of the filtration, the $\mathcal{F}_t$ -measurable random variable $W(t, x)(\omega)$, $\omega \in \Omega$, depends only on the restriction of ω to the time interval $[0, t]$.

The result of the fourth step combined with the following auxiliary Lemma 3.4 completes the proof. $\square$

LEMMA 3.4. *Let ζ be a random variable defined over our classical Wiener space $(\Omega, \mathcal{F}_T, P)$, such that $\zeta(\tau_h) = \zeta$, P -a.s., for any $h \in H$. Then $\zeta = E\zeta$, P -a.s.*

Proof. Let $h \in H$ and $A \in \mathcal{B}(\mathbb{R})$. Then

$$\begin{aligned} &E \left[\mathbf{1}_{\{\zeta \in A\}} \exp \left\{ \int_0^T \dot{h}_s dB_s - \frac{1}{2} \int_0^T |\dot{h}_s|^2 ds \right\} \right] \\ &= E \left[\mathbf{1}_{\{\zeta(\tau_{-h}) \in A\}} \exp \left\{ \int_0^T \dot{h}_s dB_s - \frac{1}{2} \int_0^T |\dot{h}_s|^2 ds \right\} \right] \\ &= E[\mathbf{1}_{\{\zeta \in A\}}], \end{aligned}$$

from which we deduce that

$$E \left[\mathbf{1}_{\{\zeta \in A\}} \exp \left\{ \int_0^T \dot{h}_s dB_s \right\} \right] = E[\mathbf{1}_{\{\zeta \in A\}}] E \left[\exp \left\{ \int_0^T \dot{h}_s dB_s \right\} \right],$$

i.e., for any $\varphi \in L^2([0, T]; \mathbb{R}^d)$,

$$(3.11) \quad E \left[\mathbf{1}_{\{\zeta \in A\}} \exp \left\{ \int_0^T \varphi_s dB_s \right\} \right] = E[\mathbf{1}_{\{\zeta \in A\}}] E \left[\exp \left\{ \int_0^T \varphi_s dB_s \right\} \right].$$

Consequently, taking into consideration the arbitrariness of $A \in \mathcal{B}(\mathbb{R})$ and of $\varphi \in L^2([0, T]; \mathbb{R}^d)$, the independence of ζ of B and hence of $\mathcal{F}_T$ follows, but this is possible only for deterministic ζ . $\square$

The first property of the lower value function $W(t, x)$ which we present is an immediate consequence of (3.6) and (3.9).

LEMMA 3.5. *There exists a constant $C > 0$ such that, for all $0 \leq t \leq T$, $x, x' \in \mathbb{R}^n$,*

$$(3.12) \quad \begin{aligned} \text{(i)} \quad &|W(t, x) - W(t, x')| \leq C|x - x'|; \\ \text{(ii)} \quad &|W(t, x)| \leq C(1 + |x|). \end{aligned}$$

We now discuss (the generalized) DPP for our stochastic differential game (3.1), (3.5), and (3.9). For this end we have to define the family of (backward) semigroups associated with BSDE (3.5). This notion of stochastic backward semigroups was first introduced by Peng [26] which was applied to study the DPP for stochastic control problems. Our approach adapts Peng's ideas to the framework of stochastic differential games.

Given the initial data (t, x) , a positive number $\delta \leq T - t$, admissible control processes $u(\cdot) \in \mathcal{U}_{t,t+\delta}$ and $v(\cdot) \in \mathcal{V}_{t,t+\delta}$, and a real-valued random variable $\eta \in L^2(\Omega, \mathcal{F}_{t+\delta}, P; \mathbb{R})$, we put

$$(3.13) \quad G_{s,t+\delta}^{t,x;u,v}[\eta] := \tilde{Y}_s^{t,x;u,v}, \quad s \in [t, t+\delta],$$

where the couple $(\tilde{Y}_s^{t,x;u,v}, \tilde{Z}_s^{t,x;u,v})_{t \leq s \leq t+\delta}$ is the solution of the following BSDE with the time horizon $t + \delta$:

$$\begin{cases} -d\tilde{Y}_s^{t,x;u,v} = f(s, X_s^{t,x;u,v}, \tilde{Y}_s^{t,x;u,v}, \tilde{Z}_s^{t,x;u,v}, u_s, v_s)ds \\ \quad -\tilde{Z}_s^{t,x;u,v} dB_s, & s \in [t, t+\delta], \\ \tilde{Y}_{t+\delta}^{t,x;u,v} = \eta, \end{cases}$$

and $X^{t,x;u,v}$ is the solution of SDE (3.1). Then, obviously, for the solution $(Y^{t,x;u,v}, Z^{t,x;u,v})$ of BSDE (3.5) we have

$$(3.14) \quad G_{t,T}^{t,x;u,v}[\Phi(X_T^{t,x;u,v})] = G_{t,t+\delta}^{t,x;u,v}[Y_{t+\delta}^{t,x;u,v}].$$

Moreover,

$$\begin{aligned} J(t, x; u, v) &= Y_{t,T}^{t,x;u,v} = G_{t,T}^{t,x;u,v}[\Phi(X_T^{t,x;u,v})] = G_{t,t+\delta}^{t,x;u,v}[Y_{t+\delta}^{t,x;u,v}] \\ &= G_{t,t+\delta}^{t,x;u,v}[J(t+\delta, X_{t+\delta}^{t,x;u,v}; u, v)]. \end{aligned}$$

Remark 3.3. When f is independent of (y, z) it holds that

$$G_{s,t+\delta}^{t,x;u,v}[\eta] = E \left[\eta + \int_s^{t+\delta} f(r, X_r^{t,x;u,v}, u_r, v_r) dr \middle| \mathcal{F}_s \right], \quad s \in [t, t+\delta].$$

THEOREM 3.6. *Under the assumptions (H3.1) and (H3.2), the lower value function $W(t, x)$ obeys the following DPP: For any $0 \leq t < t + \delta \leq T$, $x \in \mathbb{R}^n$,*

$$(3.15) \quad W(t, x) = \operatorname{essinf}_{\beta \in \mathcal{B}_{t,t+\delta}} \operatorname{esssup}_{u \in \mathcal{U}_{t,t+\delta}} G_{t,t+\delta}^{t,x;u,\beta(u)}[W(t+\delta, X_{t+\delta}^{t,x;u,\beta(u)})].$$

Proof. To simplify notations we put

$$W_\delta(t, x) = \operatorname{essinf}_{\beta \in \mathcal{B}_{t,t+\delta}} \operatorname{esssup}_{u \in \mathcal{U}_{t,t+\delta}} G_{t,t+\delta}^{t,x;u,\beta(u)}[W(t+\delta, X_{t+\delta}^{t,x;u,\beta(u)})].$$

The proof that $W_\delta(t, x)$ coincides with $W(t, x)$ will be split into the following lemmas, which all are supposed to satisfy (H3.1) and (H3.2). $\square$

LEMMA 3.7. $W_\delta(t, x)$ is deterministic.

Proof. The proof of this lemma uses the same ideas as that of Proposition 3.3, so it can be omitted here. $\square$

LEMMA 3.8. $W_\delta(t, x) \leq W(t, x)$.

Proof. Let $\beta \in \mathcal{B}_{t,T}$ be arbitrarily fixed. Then, given a $u_2(\cdot) \in \mathcal{U}_{t+\delta,T}$, we define as follows the restriction β_1 of β to $\mathcal{U}_{t,t+\delta}$:

$$\beta_1(u_1) := \beta(u_1 \oplus u_2)|_{[t,t+\delta]}, \quad u_1(\cdot) \in \mathcal{U}_{t,t+\delta},$$

where $u_1 \oplus u_2 := u_1 \mathbf{1}_{[t,t+\delta]} + u_2 \mathbf{1}_{(t+\delta,T]}$ extends $u_1(\cdot)$ to an element of $\mathcal{U}_{t,T}$. It is easy to check that $\beta_1 \in \mathcal{B}_{t,t+\delta}$. Moreover, from the nonanticipativity property of β we deduce that β_1 is independent of the special choice of $u_2(\cdot) \in \mathcal{U}_{t+\delta,T}$. Consequently, from the definition of $W_\delta(t, x)$,

$$(3.16) \quad W_\delta(t, x) \leq \text{esssup}_{u_1 \in \mathcal{U}_{t,t+\delta}} G_{t,t+\delta}^{t,x;u_1,\beta_1(u_1)}[W(t+\delta, X_{t+\delta}^{t,x;u_1,\beta_1(u_1)})], \quad P\text{-a.s.}$$

We use the notation $I_\delta(t, x, u, v) := G_{t,t+\delta}^{t,x;u,v}[W(t+\delta, X_{t+\delta}^{t,x;u,v})]$ and notice that there exists a sequence $\{u_i^1, i \geq 1\} \subset \mathcal{U}_{t,t+\delta}$ such that

$$I_\delta(t, x, \beta_1) := \text{esssup}_{u_1 \in \mathcal{U}_{t,t+\delta}} I_\delta(t, x, u_1, \beta_1(u_1)) = \sup_{i \geq 1} I_\delta(t, x, u_i^1, \beta_1(u_i^1)), \quad P\text{-a.s.}$$

For any $\varepsilon > 0$, we put $\tilde{\Gamma}_i := \{I_\delta(t, x, \beta_1) \leq I_\delta(t, x, u_i^1, \beta_1(u_i^1)) + \varepsilon\} \in \mathcal{F}_t$, $i \geq 1$. Then $\Gamma_1 := \tilde{\Gamma}_1$, $\Gamma_i := \tilde{\Gamma}_i \setminus (\cup_{l=1}^{i-1} \tilde{\Gamma}_l) \in \mathcal{F}_t$, $i \geq 2$, form an $(\Omega, \mathcal{F}_t)$ -partition, and $u_1^\varepsilon := \sum_{i \geq 1} \mathbf{1}_{\Gamma_i} u_i^1$ belongs obviously to $\mathcal{U}_{t,t+\delta}$. Moreover, from the nonanticipativity of β_1 we have $\beta_1(u_1^\varepsilon) = \sum_{i \geq 1} \mathbf{1}_{\Gamma_i} \beta_1(u_i^1)$, and from the uniqueness of the solution of the forward-backward SDE (FBSDE), we deduce that $I_\delta(t, x, u_1^\varepsilon, \beta_1(u_1^\varepsilon)) = \sum_{i \geq 1} \mathbf{1}_{\Gamma_i} I_\delta(t, x, u_i^1, \beta_1(u_i^1))$, P -a.s. Hence,

$$(3.17) \quad W_\delta(t, x) \leq I_\delta(t, x, \beta_1) \leq \sum_{i \geq 1} \mathbf{1}_{\Gamma_i} I_\delta(t, x, u_i^1, \beta_1(u_i^1)) + \varepsilon = I_\delta(t, x, u_1^\varepsilon, \beta_1(u_1^\varepsilon)) + \varepsilon \\ = G_{t,t+\delta}^{t,x;u_1^\varepsilon,\beta_1(u_1^\varepsilon)}[W(t+\delta, X_{t+\delta}^{t,x;u_1^\varepsilon,\beta_1(u_1^\varepsilon)})] + \varepsilon, \quad P\text{-a.s.}$$

On the other hand, using the fact that $\beta_1(\cdot) := \beta(\cdot \oplus u_2) \in \mathcal{B}_{t,t+\delta}$ does not depend on $u_2(\cdot) \in \mathcal{U}_{t+\delta,T}$, we can define $\beta_2(u_2) := \beta(u_1^\varepsilon \oplus u_2)|_{[t+\delta,T]}$, for all $u_2(\cdot) \in \mathcal{U}_{t+\delta,T}$. The such defined $\beta_2 : \mathcal{U}_{t+\delta,T} \rightarrow \mathcal{V}_{t+\delta,T}$ belongs to $\mathcal{B}_{t+\delta,T}$ since $\beta \in \mathcal{B}_{t,T}$. Therefore, from the definition of $W(t+\delta, y)$ we have, for any $y \in \mathbb{R}^n$,

$$W(t+\delta, y) \leq \text{esssup}_{u_2 \in \mathcal{U}_{t+\delta,T}} J(t+\delta, y; u_2, \beta_2(u_2)), \quad P\text{-a.s.}$$

Finally, because there exists a constant $C \in \mathbb{R}$ such that

$$(3.18) \quad \begin{aligned} & \text{(i)} \quad |W(t+\delta, y) - W(t+\delta, y')| \leq C|y - y'| \text{ for any } y, y' \in \mathbb{R}^n; \\ & \text{(ii)} \quad |J(t+\delta, y, u_2, \beta_2(u_2)) - J(t+\delta, y', u_2, \beta_2(u_2))| \leq C|y - y'|, \quad P\text{-a.s.,} \\ & \quad \text{for any } u_2 \in \mathcal{U}_{t+\delta,T}, \end{aligned}$$

(see Lemma 3.5(i) and (3.6)(i)) we can show by approximating $X_{t+\delta}^{t,x;u_1^\varepsilon,\beta_1(u_1^\varepsilon)}$ that

$$W(t+\delta, X_{t+\delta}^{t,x;u_1^\varepsilon,\beta_1(u_1^\varepsilon)}) \leq \text{esssup}_{u_2 \in \mathcal{U}_{t+\delta,T}} J(t+\delta, X_{t+\delta}^{t,x;u_1^\varepsilon,\beta_1(u_1^\varepsilon)}; u_2, \beta_2(u_2)), \quad P\text{-a.s.}$$

To estimate the right side of the latter inequality we note that there exists some sequence $\{u_j^2, j \geq 1\} \subset \mathcal{U}_{t+\delta,T}$ such that

$$\begin{aligned} & \text{esssup}_{u_2 \in \mathcal{U}_{t+\delta,T}} J(t+\delta, X_{t+\delta}^{t,x;u_1^\varepsilon,\beta_1(u_1^\varepsilon)}; u_2, \beta_2(u_2)) \\ & = \sup_{j \geq 1} J(t+\delta, X_{t+\delta}^{t,x;u_1^\varepsilon,\beta_1(u_1^\varepsilon)}; u_j^2, \beta_2(u_j^2)), \quad P\text{-a.s.} \end{aligned}$$

Then, putting $\tilde{\Delta}_j := \{\text{esssup}_{u_2 \in \mathcal{U}_{t+\delta, T}} J(t + \delta, X_{t+\delta}^{t, x; u_1^\varepsilon, \beta_1(u_1^\varepsilon)}; u_2, \beta_2(u_2)) \leq J(t + \delta, X_{t+\delta}^{t, x; u_1^\varepsilon, \beta_1(u_1^\varepsilon)}; u_j^2, \beta_2(u_j^2)) + \varepsilon\} \in \mathcal{F}_{t+\delta}$, $j \geq 1$; we have with $\Delta_1 := \tilde{\Delta}_1$, $\Delta_j := \tilde{\Delta}_j \setminus (\cup_{l=1}^{j-1} \tilde{\Delta}_l) \in \mathcal{F}_{t+\delta}$, $j \geq 2$, an $(\Omega, \mathcal{F}_{t+\delta})$ -partition and $u_2^\varepsilon := \sum_{j \geq 1} \mathbf{1}_{\Delta_j} u_j^2 \in \mathcal{U}_{t+\delta, T}$. From the nonanticipativity of β_2 we have $\beta_2(u_2^\varepsilon) = \sum_{j \geq 1} \mathbf{1}_{\Delta_j} \beta_2(u_j^2)$, and from the definition of β_1 , β_2 we know that $\beta(u_1^\varepsilon \oplus u_2^\varepsilon) = \beta_1(u_1^\varepsilon) \oplus \beta_2(u_2^\varepsilon)$. Thus, again from the uniqueness of the solution of our FBSDE, we get

$$\begin{aligned} J(t + \delta, X_{t+\delta}^{t, x; u_1^\varepsilon, \beta_1(u_1^\varepsilon)}; u_2^\varepsilon, \beta_2(u_2^\varepsilon)) &= Y_{t+\delta}^{t+\delta, X_{t+\delta}^{t, x; u_1^\varepsilon, \beta_1(u_1^\varepsilon)}; u_2^\varepsilon, \beta_2(u_2^\varepsilon)} \quad (\text{see (3.8)}) \\ &= \sum_{j \geq 1} \mathbf{1}_{\Delta_j} Y_{t+\delta}^{t+\delta, X_{t+\delta}^{t, x; u_1^\varepsilon, \beta_1(u_1^\varepsilon)}; u_j^2, \beta_2(u_j^2)} \\ &= \sum_{j \geq 1} \mathbf{1}_{\Delta_j} J(t + \delta, X_{t+\delta}^{t, x; u_1^\varepsilon, \beta_1(u_1^\varepsilon)}; u_j^2, \beta_2(u_j^2)), \quad P\text{-a.s.} \end{aligned}$$

Consequently,

$$\begin{aligned} (3.19) \quad W(t + \delta, X_{t+\delta}^{t, x; u_1^\varepsilon, \beta_1(u_1^\varepsilon)}) &\leq \text{esssup}_{u_2 \in \mathcal{U}_{t+\delta, T}} J(t + \delta, X_{t+\delta}^{t, x; u_1^\varepsilon, \beta_1(u_1^\varepsilon)}; u_2, \beta_2(u_2)) \\ &\leq \sum_{j \geq 1} \mathbf{1}_{\Delta_j} Y_{t+\delta}^{t, x; u_1^\varepsilon \oplus u_j^2, \beta(u_1^\varepsilon \oplus u_j^2)} + \varepsilon \\ &= Y_{t+\delta}^{t, x; u_1^\varepsilon \oplus u_2^\varepsilon, \beta(u_1^\varepsilon \oplus u_2^\varepsilon)} + \varepsilon \\ &= Y_{t+\delta}^{t, x; u^\varepsilon, \beta(u^\varepsilon)} + \varepsilon, \quad P\text{-a.s.}, \end{aligned}$$

where $u^\varepsilon := u_1^\varepsilon \oplus u_2^\varepsilon \in \mathcal{U}_{t, T}$. From (3.17) and (3.19) and Lemmas 2.2 (comparison theorem for BSDEs) and 2.3, we have

$$\begin{aligned} (3.20) \quad W_\delta(t, x) &\leq G_{t, t+\delta}^{t, x; u_1^\varepsilon, \beta_1(u_1^\varepsilon)} [Y_{t+\delta}^{t, x; u^\varepsilon, \beta(u^\varepsilon)} + \varepsilon] + \varepsilon \\ &\leq G_{t, t+\delta}^{t, x; u_1^\varepsilon, \beta_1(u_1^\varepsilon)} [Y_{t+\delta}^{t, x; u^\varepsilon, \beta(u^\varepsilon)}] + (C + 1)\varepsilon \\ &= G_{t, t+\delta}^{t, x; u^\varepsilon, \beta(u^\varepsilon)} [Y_{t+\delta}^{t, x; u^\varepsilon, \beta(u^\varepsilon)}] + (C + 1)\varepsilon \\ &= Y_t^{t, x; u^\varepsilon, \beta(u^\varepsilon)} + (C + 1)\varepsilon \\ &\leq \text{esssup}_{u \in \mathcal{U}_{t, T}} Y_t^{t, x; u, \beta(u)} + (C + 1)\varepsilon, \quad P\text{-a.s.} \end{aligned}$$

Since $\beta \in \mathcal{B}_{t, T}$ has been arbitrarily chosen we have (3.20) for all $\beta \in \mathcal{B}_{t, T}$. Therefore,

$$(3.21) \quad W_\delta(t, x) \leq \text{essinf}_{\beta \in \mathcal{B}_{t, T}} \text{esssup}_{u \in \mathcal{U}_{t, T}} Y_t^{t, x; u, \beta(u)} + (C + 1)\varepsilon = W(t, x) + (C + 1)\varepsilon.$$

Finally, letting $\varepsilon \downarrow 0$, we get $W_\delta(t, x) \leq W(t, x)$. $\square$

LEMMA 3.9. $W(t, x) \leq W_\delta(t, x)$.

Proof. We continue to use the notations introduced above, and from the definition of $W_\delta(t, x)$ we have

$$\begin{aligned} W_\delta(t, x) &= \text{essinf}_{\beta_1 \in \mathcal{B}_{t, t+\delta}} \text{esssup}_{u_1 \in \mathcal{U}_{t, t+\delta}} G_{t, t+\delta}^{t, x; u_1, \beta_1(u_1)} [W(t + \delta, X_{t+\delta}^{t, x; u_1, \beta_1(u_1)})] \\ &= \text{essinf}_{\beta_1 \in \mathcal{B}_{t, t+\delta}} I_\delta(t, x, \beta_1), \end{aligned}$$

and, for some sequence $\{\beta_i^1, i \geq 1\} \subset \mathcal{B}_{t, t+\delta}$,

$$W_\delta(t, x) = \inf_{i \geq 1} I_\delta(t, x, \beta_i^1), \quad P\text{-a.s.}$$

For any $\varepsilon > 0$, we let $\tilde{\Lambda}_i := \{I_\delta(t, x, \beta_i^1) - \varepsilon \leq W_\delta(t, x)\} \in \mathcal{F}_t$, $i \geq 1$, $\Lambda_1 := \tilde{\Lambda}_1$ and $\Lambda_i := \tilde{\Lambda}_i \setminus (\cup_{l=1}^{i-1} \tilde{\Lambda}_l) \in \mathcal{F}_t$, $i \geq 2$. Then $\{\Lambda_i, i \geq 1\}$ is an $(\Omega, \mathcal{F}_t)$ -partition, $\beta_1^\varepsilon := \sum_{i \geq 1} \mathbf{1}_{\Lambda_i} \beta_i^1$ belongs to $\mathcal{B}_{t,t+\delta}$, and from the uniqueness of the solution of our FBSDE we conclude that $I_\delta(t, x, u_1, \beta_1^\varepsilon(u_1)) = \sum_{i \geq 1} \mathbf{1}_{\Lambda_i} I_\delta(t, x, u_1, \beta_i^1(u_1))$, P -a.s., for all $u_1(\cdot) \in \mathcal{U}_{t,t+\delta}$. Hence,

(3.22)

$$\begin{aligned} W_\delta(t, x) &\geq \sum_{i \geq 1} \mathbf{1}_{\Lambda_i} I_\delta(t, x, \beta_i^1) - \varepsilon \\ &\geq \sum_{i \geq 1} \mathbf{1}_{\Lambda_i} I_\delta(t, x, u_1, \beta_i^1(u_1)) - \varepsilon \\ &= I_\delta(t, x, u_1, \beta_1^\varepsilon(u_1)) - \varepsilon \\ &= G_{t,t+\delta}^{t,x;u_1,\beta_1^\varepsilon(u_1)}[W(t+\delta, X_{t+\delta}^{t,x;u_1,\beta_1^\varepsilon(u_1)})] - \varepsilon, \quad P\text{-a.s.}, \text{ for all } u_1 \in \mathcal{U}_{t,t+\delta}. \end{aligned}$$

On the other hand, from the definition of $W(t+\delta, y)$, with the same technique as before, we deduce that, for any $y \in \mathbb{R}^n$, there exists $\beta_y^\varepsilon \in \mathcal{B}_{t+\delta,T}$ such that

$$(3.23) \quad W(t+\delta, y) \geq \text{esssup}_{u_2 \in \mathcal{U}_{t+\delta,T}} J(t+\delta, y; u_2, \beta_y^\varepsilon(u_2)) - \varepsilon, \quad P\text{-a.s.}$$

Let $\{O_i\}_{i \geq 1} \subset \mathcal{B}(\mathbb{R}^n)$ be a decomposition of $\mathbb{R}^n$ such that $\sum_{i \geq 1} O_i = \mathbb{R}^n$ and $\text{diam}(O_i) \leq \varepsilon$, $i \geq 1$. Let y_i be an arbitrarily fixed element of O_i , $i \geq 1$. Defining $[X_{t+\delta}^{t,x;u_1,\beta_1^\varepsilon(u_1)}] := \sum_{i \geq 1} y_i \mathbf{1}_{\{X_{t+\delta}^{t,x;u_1,\beta_1^\varepsilon(u_1)} \in O_i\}}$, we have

$$(3.24) \quad |X_{t+\delta}^{t,x;u_1,\beta_1^\varepsilon(u_1)} - [X_{t+\delta}^{t,x;u_1,\beta_1^\varepsilon(u_1)}]| \leq \varepsilon, \text{ everywhere on } \Omega, \text{ for all } u_1 \in \mathcal{U}_{t,t+\delta}.$$

Moreover, for each y_i , there exists some $\beta_{y_i}^\varepsilon \in \mathcal{B}_{t+\delta,T}$ such that (3.23) holds, and, clearly, $\beta_{u_1}^\varepsilon := \sum_{i \geq 1} \mathbf{1}_{\{X_{t+\delta}^{t,x;u_1,\beta_1^\varepsilon(u_1)} \in O_i\}} \beta_{y_i}^\varepsilon \in \mathcal{B}_{t+\delta,T}$.

Now we can define the new strategy $\beta^\varepsilon(u) := \beta_1^\varepsilon(u_1) \oplus \beta_{u_1}^\varepsilon(u_2)$, $u \in \mathcal{U}_{t,T}$, where $u_1 = u|_{[t,t+\delta]}$, $u_2 = u|_{(t+\delta,T]}$ (restriction of u to $[t,t+\delta] \times \Omega$ and $(t+\delta,T] \times \Omega$, resp.). Obviously, β^ε maps $\mathcal{U}_{t,T}$ into $\mathcal{V}_{t,T}$. Moreover, β^ε is nonanticipating: Indeed, let $S : \Omega \rightarrow [t,T]$ be an $\mathbb{F}$ -stopping time and $u, u' \in \mathcal{U}_{t,T}$ be such that $u \equiv u'$ on $\llbracket t, S \rrbracket$. Decomposing u, u' into $u_1, u'_1 \in \mathcal{U}_{t,t+\delta}$, $u_2, u'_2 \in \mathcal{U}_{t+\delta,T}$ such that $u = u_1 \oplus u_2$ and $u' = u'_1 \oplus u'_2$. We have $u_1 \equiv u'_1$ on $\llbracket t, S \wedge (t+\delta) \rrbracket$ from which we get $\beta_1^\varepsilon(u_1) \equiv \beta_1^\varepsilon(u'_1)$ on $\llbracket t, S \wedge (t+\delta) \rrbracket$ (recall that β_1^ε is nonanticipating). On the other hand, $u_2 \equiv u'_2$ on $\llbracket t+\delta, S \vee (t+\delta) \rrbracket \subset (t+\delta, T] \times \{S > t+\delta\}$, and on $\{S > t+\delta\}$ we have $X_{t+\delta}^{t,x;u_1,\beta_1^\varepsilon(u_1)} = X_{t+\delta}^{t,x;u'_1,\beta_1^\varepsilon(u'_1)}$. Consequently, from our definition, $\beta_{u_1}^\varepsilon = \beta_{u'_1}^\varepsilon$ on $\{S > t+\delta\}$ and $\beta_{u_1}^\varepsilon(u_2) \equiv \beta_{u'_1}^\varepsilon(u'_2)$ on $\llbracket t+\delta, S \vee (t+\delta) \rrbracket$. This yields $\beta^\varepsilon(u) = \beta_1^\varepsilon(u_1) \oplus \beta_{u_1}^\varepsilon(u_2) \equiv \beta_1^\varepsilon(u'_1) \oplus \beta_{u'_1}^\varepsilon(u'_2) = \beta^\varepsilon(u')$ on $\llbracket t, S \rrbracket$, from which it follows that $\beta^\varepsilon \in \mathcal{B}_{t,T}$.

Let now $u \in \mathcal{U}_{t,T}$ be arbitrarily chosen and decomposed into $u_1 = u|_{[t,t+\delta]} \in \mathcal{U}_{t,t+\delta}$ and $u_2 = u|_{(t+\delta,T]} \in \mathcal{U}_{t+\delta,T}$. Then, from (3.22), (3.18)(i), (3.24), and Lemmas 2.2 (comparison theorem) and 2.3, we obtain

(3.25)

$$\begin{aligned} W_\delta(t, x) &\geq G_{t,t+\delta}^{t,x;u_1,\beta_1^\varepsilon(u_1)}[W(t+\delta, X_{t+\delta}^{t,x;u_1,\beta_1^\varepsilon(u_1)})] - \varepsilon \\ &\geq G_{t,t+\delta}^{t,x;u_1,\beta_1^\varepsilon(u_1)}[W(t+\delta, [X_{t+\delta}^{t,x;u_1,\beta_1^\varepsilon(u_1)}]) - C\varepsilon] - \varepsilon \\ &\geq G_{t,t+\delta}^{t,x;u_1,\beta_1^\varepsilon(u_1)}[W(t+\delta, [X_{t+\delta}^{t,x;u_1,\beta_1^\varepsilon(u_1)}])] - C\varepsilon \end{aligned}$$

$$= G_{t,t+\delta}^{t,x;u_1,\beta_1^\varepsilon(u_1)} \left[\sum_{i \geq 1} \mathbf{1}_{\{X_{t+\delta}^{t,x;u_1,\beta_1^\varepsilon(u_1)} \in O_i\}} W(t+\delta, y_i) \right] - C\varepsilon, \quad P\text{-a.s.}$$

Furthermore, from (3.23), (3.18)(ii), (3.24), and Lemmas 2.2 and 2.3, we have

(3.26)

$$\begin{aligned} W_\delta(t, x) &\geq G_{t,t+\delta}^{t,x;u_1,\beta_1^\varepsilon(u_1)} \left[\sum_{i \geq 1} \mathbf{1}_{\{X_{t+\delta}^{t,x;u_1,\beta_1^\varepsilon(u_1)} \in O_i\}} J(t+\delta, y_i; u_2, \beta_{y_i}^\varepsilon(u_2)) - \varepsilon \right] - C\varepsilon \\ &\geq G_{t,t+\delta}^{t,x;u_1,\beta_1^\varepsilon(u_1)} \left[\sum_{i \geq 1} \mathbf{1}_{\{X_{t+\delta}^{t,x;u_1,\beta_1^\varepsilon(u_1)} \in O_i\}} J(t+\delta, y_i; u_2, \beta_{y_i}^\varepsilon(u_2)) \right] - C\varepsilon \\ &= G_{t,t+\delta}^{t,x;u_1,\beta_1^\varepsilon(u_1)} \left[J(t+\delta, [X_{t+\delta}^{t,x;u_1,\beta_1^\varepsilon(u_1)}]; u_2, \beta_{u_1}^\varepsilon(u_2)) \right] - C\varepsilon \\ &\geq G_{t,t+\delta}^{t,x;u_1,\beta_1^\varepsilon(u_1)} [J(t+\delta, X_{t+\delta}^{t,x;u_1,\beta_1^\varepsilon(u_1)}; u_2, \beta_{u_1}^\varepsilon(u_2)) - C\varepsilon] - C\varepsilon \\ &\geq G_{t,t+\delta}^{t,x;u_1,\beta_1^\varepsilon(u_1)} [J(t+\delta, X_{t+\delta}^{t,x;u_1,\beta_1^\varepsilon(u_1)}; u_2, \beta_{u_1}^\varepsilon(u_2))] - C\varepsilon \\ &= G_{t,t+\delta}^{t,x;u,\beta^\varepsilon(u)} [Y_{t+\delta}^{t,x,u,\beta^\varepsilon(u)}] - C\varepsilon \\ &= Y_t^{t,x;u,\beta^\varepsilon(u)} - C\varepsilon, \quad P\text{-a.s., for any } u \in \mathcal{U}_{t,T}. \end{aligned}$$

Consequently,

$$\begin{aligned} W_\delta(t, x) &\geq \text{esssup}_{u \in \mathcal{U}_{t,T}} J(t, x; u, \beta^\varepsilon(u)) - C\varepsilon \\ (3.27) \quad &\geq \text{essinf}_{\beta \in \mathcal{B}_{t,T}} \text{esssup}_{u \in \mathcal{U}_{t,T}} J(t, x; u, \beta(u)) - C\varepsilon \\ &= W(t, x) - C\varepsilon, \quad P\text{-a.s.} \end{aligned}$$

Finally, letting $\varepsilon \downarrow 0$ we get $W_\delta(t, x) \geq W(t, x)$. The proof is complete. $\square$

Remark 3.4. (i) From the inequalities (3.17) and (3.22) we see that for all $(t, x) \in [0, T] \times \mathbb{R}^n$, $\delta > 0$, with $0 < \delta \leq T - t$ and $\varepsilon > 0$, the following hold: (a) For every $\beta \in \mathcal{B}_{t,t+\delta}$, there exists some $u^\varepsilon(\cdot) \in \mathcal{U}_{t,t+\delta}$ such that

$$(3.28) \quad W(t, x) (= W_\delta(t, x)) \leq G_{t,t+\delta}^{t,x;u^\varepsilon,\beta(u^\varepsilon)} [W(t+\delta, X_{t+\delta}^{t,x;u^\varepsilon,\beta(u^\varepsilon)})] + \varepsilon, \quad P\text{-a.s.}$$

(b) There exists some $\beta^\varepsilon \in \mathcal{B}_{t,t+\delta}$ such that, for all $u \in \mathcal{U}_{t,t+\delta}$,

$$(3.29) \quad W(t, x) (= W_\delta(t, x)) \geq G_{t,t+\delta}^{t,x;u,\beta^\varepsilon(u)} [W(t+\delta, X_{t+\delta}^{t,x;u,\beta^\varepsilon(u)})] - \varepsilon, \quad P\text{-a.s.}$$

(ii) Recall that the lower value function W is deterministic. Thus, by choosing $\delta = T - t$ and taking the expectation on both sides of (3.28) and (3.29) we can show that

$$W(t, x) = \inf_{\beta \in \mathcal{B}_{t,T}} \sup_{u \in \mathcal{U}_{t,T}} E[J(t, x; u, \beta(u))].$$

In analogy we also have

$$U(t, x) = \sup_{\alpha \in \mathcal{A}_{t,T}} \inf_{v \in \mathcal{V}_{t,T}} E[J(t, x; \alpha(v), v)].$$

The above formulas look similar to the definitions of the lower and the upper value functions defined by Fleming and Souganidis [14] for the case of f being independent

of (y, z) . However, they consider only control processes which are independent of the past $\mathcal{F}_t$. In Remark 5.3 we will come back to this comparison and identify their value functions with ours for such a coefficient f .

In Lemma 3.5 we have already seen that the lower value function $W(t, x)$ is Lipschitz continuous in x , uniformly in t . With the help of Theorem 3.6 we can now also study the continuity property of $W(t, x)$ in t .

THEOREM 3.10. *Let us suppose that the assumptions (H3.1) and (H3.2) hold. Then the lower value function $W(t, x)$ is $\frac{1}{2}$ -Hölder continuous in t : There exists a constant C such that, for every $x \in \mathbb{R}^n$, $t, t' \in [0, T]$,*

$$|W(t, x) - W(t', x)| \leq C(1 + |x|)|t - t'|^{\frac{1}{2}}.$$

Proof. Let $(t, x) \in [0, T] \times \mathbb{R}^n$ and $\delta > 0$ be arbitrarily given such that $0 < \delta \leq T - t$. Our objective is to prove the following inequality by using (3.28) and (3.29):

$$(3.30) \quad -C(1 + |x|)\delta^{\frac{1}{2}} \leq W(t, x) - W(t + \delta, x) \leq C(1 + |x|)\delta^{\frac{1}{2}}.$$

From it we obtain immediately that W is $\frac{1}{2}$ -Hölder continuous in t . We will check only the second inequality in (3.30); the first one can be shown in a similar way. To this end we note that due to (3.28), for an arbitrarily small $\varepsilon > 0$,

$$(3.31) \quad W(t, x) - W(t + \delta, x) \leq I_\delta^1 + I_\delta^2 + \varepsilon,$$

where

$$\begin{aligned} I_\delta^1 &:= G_{t, t+\delta}^{t, x; u^\varepsilon, \beta(u^\varepsilon)}[W(t + \delta, X_{t+\delta}^{t, x; u^\varepsilon, \beta(u^\varepsilon)})] - G_{t, t+\delta}^{t, x; u^\varepsilon, \beta(u^\varepsilon)}[W(t + \delta, x)], \\ I_\delta^2 &:= G_{t, t+\delta}^{t, x; u^\varepsilon, \beta(u^\varepsilon)}[W(t + \delta, x)] - W(t + \delta, x) \end{aligned}$$

for arbitrarily chosen $\beta \in \mathcal{B}_{t, t+\delta}$ and $u^\varepsilon \in \mathcal{U}_{t, t+\delta}$ such that (3.28) holds. From Lemma 2.3 and the estimate (3.12) we obtain that, for some constant C independent of the controls u^ε and $\beta(u^\varepsilon)$,

$$\begin{aligned} |I_\delta^1| &\leq [CE(|W(t + \delta, X_{t+\delta}^{t, x; u^\varepsilon, \beta(u^\varepsilon)}) - W(t + \delta, x)|^2 | \mathcal{F}_t)]^{\frac{1}{2}} \\ &\leq [CE(|X_{t+\delta}^{t, x; u^\varepsilon, \beta(u^\varepsilon)} - x|^2 | \mathcal{F}_t)]^{\frac{1}{2}}, \end{aligned}$$

and since $E[|X_{t+\delta}^{t, x; u^\varepsilon, \beta(u^\varepsilon)} - x|^2 | \mathcal{F}_t] \leq C(1 + |x|^2)\delta$ we deduce that $|I_\delta^1| \leq C(1 + |x|)\delta^{\frac{1}{2}}$. From the definition of $G_{t, t+\delta}^{t, x; u^\varepsilon, \beta(u^\varepsilon)}[\cdot]$ (see (3.13)) we know that the second term I_δ^2 can be written as

$$\begin{aligned} I_\delta^2 &= E \left[W(t + \delta, x) + \int_t^{t+\delta} f(s, X_s^{t, x; u^\varepsilon, \beta(u^\varepsilon)}, \tilde{Y}_s^{t, x; u^\varepsilon, \beta(u^\varepsilon)}, \tilde{Z}_s^{t, x; u^\varepsilon, \beta(u^\varepsilon)}, u_s^\varepsilon, \beta_s(u^\varepsilon)) ds \right. \\ &\quad \left. - \int_t^{t+\delta} \tilde{Z}_s^{t, x; u^\varepsilon, \beta(u^\varepsilon)} dB_s | \mathcal{F}_t \right] - W(t + \delta, x) \\ &= E \left[\int_t^{t+\delta} f(s, X_s^{t, x; u^\varepsilon, \beta(u^\varepsilon)}, \tilde{Y}_s^{t, x; u^\varepsilon, \beta(u^\varepsilon)}, \tilde{Z}_s^{t, x; u^\varepsilon, \beta(u^\varepsilon)}, u_s^\varepsilon, \beta_s(u^\varepsilon)) ds | \mathcal{F}_t \right]. \end{aligned}$$

With the help of the Schwartz inequality and the estimates (3.3) and (A.4)(i), we

then have

$$\begin{aligned}
|I_\delta^2| &\leq \delta^{\frac{1}{2}} E \left[\int_t^{t+\delta} |f(s, X_s^{t,x;u^\varepsilon, \beta(u^\varepsilon)}, \tilde{Y}_s^{t,x;u^\varepsilon, \beta(u^\varepsilon)}, \tilde{Z}_s^{t,x;u^\varepsilon, \beta(u^\varepsilon)}, u_s^\varepsilon, \beta_s(u^\varepsilon))|^2 ds | \mathcal{F}_t \right]^{\frac{1}{2}} \\
&\leq \delta^{\frac{1}{2}} E \left[\int_t^{t+\delta} (|f(s, X_s^{t,x;u^\varepsilon, \beta(u^\varepsilon)}, 0, 0, u_s^\varepsilon, \beta_s(u^\varepsilon))| + C|\tilde{Y}_s^{t,x;u^\varepsilon, \beta(u^\varepsilon)}| \right. \\
&\quad \left. + C|\tilde{Z}_s^{t,x;u^\varepsilon, \beta(u^\varepsilon)}|)^2 ds | \mathcal{F}_t \right]^{\frac{1}{2}} \\
&\leq C\delta^{\frac{1}{2}} E \left[\int_t^{t+\delta} (1 + |X_s^{t,x;u^\varepsilon, \beta(u^\varepsilon)}| + |\tilde{Y}_s^{t,x;u^\varepsilon, \beta(u^\varepsilon)}| + |\tilde{Z}_s^{t,x;u^\varepsilon, \beta(u^\varepsilon)}|)^2 ds | \mathcal{F}_t \right]^{\frac{1}{2}} \\
&\leq C(1 + |x|)\delta^{\frac{1}{2}}.
\end{aligned}$$

Hence, from (3.31),

$$W(t, x) - W(t + \delta, x) \leq C(1 + |x|)\delta^{\frac{1}{2}} + \varepsilon,$$

and letting $\varepsilon \downarrow 0$ we get the second inequality of (3.30). The proof is complete. $\square$

4. Viscosity solution of Isaacs' equation: Existence theorem. In this section we consider the following Isaacs' equations:

$$(4.1) \quad \begin{cases} \frac{\partial}{\partial t} W(t, x) + H^-(t, x, W, DW, D^2W) = 0, & (t, x) \in [0, T) \times \mathbb{R}^n, \\ W(T, x) = \Phi(x), & x \in \mathbb{R}^n, \end{cases}$$

and

$$(4.2) \quad \begin{cases} \frac{\partial}{\partial t} U(t, x) + H^+(t, x, U, DU, D^2U) = 0, & (t, x) \in [0, T) \times \mathbb{R}^n, \\ U(T, x) = \Phi(x), & x \in \mathbb{R}^n, \end{cases}$$

associated with the Hamiltonians

$$H^-(t, x, y, p, X) = \sup_{u \in U} \inf_{v \in V} \left\{ \frac{1}{2} \text{tr}(\sigma \sigma^T(t, x, u, v) X) + p \cdot b(t, x, u, v) + f(t, x, y, p \cdot \sigma, u, v) \right\}$$

and

$$H^+(t, x, y, p, X) = \inf_{v \in V} \sup_{u \in U} \left\{ \frac{1}{2} \text{tr}(\sigma \sigma^T(t, x, u, v) X) + p \cdot b(t, x, u, v) + f(t, x, y, p \cdot \sigma, u, v) \right\},$$

respectively, where $t \in [0, T]$, $x \in \mathbb{R}^n$, $y \in \mathbb{R}$, $p \in \mathbb{R}^n$, and $X \in \mathbf{S}^n$ (recall that $\mathbf{S}^n$ denotes the set of $n \times n$ symmetric matrices). Here the functions b, σ, f , and Φ are supposed to satisfy (H3.1) and (H3.2), respectively.

In this section we want to prove that the lower value function $W(t, x)$ introduced by (3.9) is the viscosity solution of (4.1), while the upper value function $U(t, x)$ defined by (3.10) is the viscosity solution of (4.2). For this we extend Peng's BSDE approach [26] developed in the framework of stochastic control theory into that of stochastic differential games. The difficulties related with this extension come from the fact that now, contrarily to the framework of stochastic control theory studied by Peng, we have to do with stochastic differential games in which strategies are played versus controls. In order to overcome these difficulties in the proof that W is a viscosity supersolution, we have, in particular, to enrich Peng's BSDE method by Lemma

4.6 and to adapt heavily the proof of Lemma 4.5. On the other hand, the proof that W is a viscosity subsolution is not covered by Peng's BSDE method and requires a quite new approach. The uniqueness of the viscosity solution will be shown in the next section for the class of continuous functions satisfying some growth assumption which is weaker than the polynomial growth condition. We first recall the definition of a viscosity solution of (4.1) and similarly for (4.2). The reader more interested in viscosity solutions is referred to Crandall, Ishii, and Lions [8].

DEFINITION 4.1. A real-valued continuous function $W \in C([0, T] \times \mathbb{R}^n)$ is called
(i) a viscosity subsolution of (4.1) if $W(T, x) \leq \Phi(x)$ for all $x \in \mathbb{R}^n$ and if for all functions $\varphi \in C_{l,b}^3([0, T] \times \mathbb{R}^n)$ and $(t, x) \in [0, T] \times \mathbb{R}^n$ such that $W - \varphi$ attains its local maximum at (t, x) :

$$\frac{\partial \varphi}{\partial t}(t, x) + H^-(t, x, \varphi, D\varphi, D^2\varphi) \geq 0;$$

(ii) a viscosity supersolution of (4.1) if $W(T, x) \geq \Phi(x)$ for all $x \in \mathbb{R}^n$ and if for all functions $\varphi \in C_{l,b}^3([0, T] \times \mathbb{R}^n)$ and $(t, x) \in [0, T] \times \mathbb{R}^n$ such that $W - \varphi$ attains its local minimum at (t, x) :

$$\frac{\partial \varphi}{\partial t}(t, x) + H^-(t, x, \varphi, D\varphi, D^2\varphi) \leq 0;$$

(iii) a viscosity solution of (4.1) if it is both a viscosity sub- and a supersolution of (4.1).

Remark 4.1. $C_{l,b}^3([0, T] \times \mathbb{R}^n)$ denotes the set of the real-valued functions that are continuously differentiable up to the third order and whose derivatives of order from 1 to 3 are bounded.

We first prove that the lower value function $W(t, x)$ is a viscosity solution of (4.1).

THEOREM 4.2. Under the assumptions (H3.1) and (H3.2) the lower value function $W(t, x)$ is a viscosity solution of (4.1).

For the proof of this theorem we need four auxiliary lemmas. To abbreviate notations we put, for some arbitrarily chosen but fixed $\varphi \in C_{l,b}^3([0, T] \times \mathbb{R}^n)$,

$$(4.3) \quad F(s, x, y, z, u, v) = \frac{\partial}{\partial s} \varphi(s, x) + \frac{1}{2} \text{tr}(\sigma \sigma^T(s, x, u, v) D^2 \varphi) + D\varphi \cdot b(s, x, u, v) \\ + f(s, x, y + \varphi(s, x), z + D\varphi(s, x) \cdot \sigma(s, x, u, v), u, v),$$

$(s, x, y, z, u, v) \in [0, T] \times \mathbb{R}^n \times \mathbb{R} \times \mathbb{R}^d \times U \times V$, and we consider the following BSDE defined on the interval $[t, t + \delta]$ ($0 < \delta \leq T - t$):

$$(4.4) \quad \begin{cases} -dY_s^{1,u,v} = F(s, X_s^{t,x;u,v}, Y_s^{1,u,v}, Z_s^{1,u,v}, u_s, v_s) ds - Z_s^{1,u,v} dB_s, \\ Y_{t+\delta}^{1,u,v} = 0, \end{cases}$$

where the process $X^{t,x,u,v}$ has been introduced by (3.1) and $u(\cdot) \in \mathcal{U}_{t,t+\delta}$, $v(\cdot) \in \mathcal{V}_{t,t+\delta}$.

Remark 4.2. It is not hard to check that $F(s, X_s^{t,x;u,v}, y, z, u_s, v_s)$ satisfies (A1) and (A2). Thus, due to Lemma 2.1, (4.4) has a unique solution.

We can characterize the solution process $Y^{1,u,v}$ as follows.

LEMMA 4.3. For every $s \in [t, t + \delta]$, we have the following relationship:

$$(4.5) \quad Y_s^{1,u,v} = G_{s,t+\delta}^{t,x;u,v}[\varphi(t + \delta, X_{t+\delta}^{t,x;u,v})] - \varphi(s, X_s^{t,x;u,v}), \quad P\text{-a.s.}$$

Proof. We recall that $G_{s,t+\delta}^{t,x;u,v}[\varphi(t+\delta, X_{t+\delta}^{t,x;u,v})]$ is defined with the help of the solution of the BSDE

$$\begin{cases} -dY_s^{u,v} = f(s, X_s^{t,x;u,v}, Y_s^{u,v}, Z_s^{u,v}, u_s, v_s)ds - Z_s^{u,v}dB_s, & s \in [t, t+\delta], \\ Y_{t+\delta}^{u,v} = \varphi(t+\delta, X_{t+\delta}^{t,x;u,v}), \end{cases}$$

by the following formula:

$$(4.6) \quad G_{s,t+\delta}^{t,x;u,v}[\varphi(t+\delta, X_{t+\delta}^{t,x;u,v})] = Y_s^{u,v}, \quad s \in [t, t+\delta]$$

(see (3.13)). Therefore we only need to prove that $Y_s^{u,v} - \varphi(s, X_s^{t,x;u,v}) \equiv Y_s^{1,u,v}$. This result can be obtained easily by applying Itô's formula to $\varphi(s, X_s^{t,x;u,v})$. Indeed, we get that the stochastic differentials of $Y_s^{u,v} - \varphi(s, X_s^{t,x;u,v})$ and $Y_s^{1,u,v}$ coincide, while at the terminal time $t+\delta$, $Y_{t+\delta}^{u,v} - \varphi(t+\delta, X_{t+\delta}^{t,x;u,v}) = 0 = Y_{t+\delta}^{1,u,v}$. So the proof is complete. $\square$

Now we consider the following simple BSDE in which the driving process $X^{t,x;u,v}$ is replaced by its deterministic initial value x :

$$(4.7) \quad \begin{cases} -dY_s^{2,u,v} = F(s, x, Y_s^{2,u,v}, Z_s^{2,u,v}, u_s, v_s)ds - Z_s^{2,u,v}dB_s, \\ Y_{t+\delta}^{2,u,v} = 0, \quad s \in [t, t+\delta], \end{cases}$$

where $u(\cdot) \in \mathcal{U}_{t,t+\delta}$, $v(\cdot) \in \mathcal{V}_{t,t+\delta}$. The following lemma will allow us to neglect the difference $|Y_t^{1,u,v} - Y_t^{2,u,v}|$ for sufficiently small $\delta > 0$.

LEMMA 4.4. *For every $u \in \mathcal{U}_{t,t+\delta}$, $v \in \mathcal{V}_{t,t+\delta}$, we have*

$$(4.8) \quad |Y_t^{1,u,v} - Y_t^{2,u,v}| \leq C\delta^{\frac{3}{2}}, \quad P\text{-a.s.},$$

where C is independent of the control processes u and v .

Proof. From (3.3) we have for all $p \geq 2$ the existence of some $C_p \in \mathbb{R}_+$ such that

$$E \left[\sup_{t \leq s \leq T} |X_s^{t,x;u,v}|^p | \mathcal{F}_t \right] \leq C_p(1 + |x|^p), \quad P\text{-a.s.}, \quad \text{uniformly in } u \in \mathcal{U}_{t,t+\delta}, v \in \mathcal{V}_{t,t+\delta}.$$

This combined with the estimate

$$\begin{aligned} E \left[\sup_{t \leq s \leq t+\delta} |X_s^{t,x;u,v} - x|^p | \mathcal{F}_t \right] &\leq 2^{p-1} E \left[\sup_{t \leq s \leq t+\delta} \left| \int_t^s b(r, X_r^{t,x;u,v}, u_r, v_r) dr \right|^p | \mathcal{F}_t \right] \\ &\quad + 2^{p-1} E \left[\sup_{t \leq s \leq t+\delta} \left| \int_t^s \sigma(r, X_r^{t,x;u,v}, u_r, v_r) dB_r \right|^p | \mathcal{F}_t \right] \end{aligned}$$

yields

$$(4.9) \quad E \left[\sup_{t \leq s \leq t+\delta} |X_s^{t,x;u,v} - x|^p | \mathcal{F}_t \right] \leq C_p \delta^{\frac{p}{2}}, \quad P\text{-a.s.}, \quad \text{uniformly in } u \in \mathcal{U}_{t,t+\delta}, v \in \mathcal{V}_{t,t+\delta}.$$

We now apply Lemma 2.3 combined with (4.9) to (4.4) and (4.7). For this we set in Lemma 2.3:

$$\xi_1 = \xi_2 = 0, \quad g(s, y, z) = F(s, X_s^{t,x;u,v}, y, z, u_s, v_s),$$

$$\varphi_1(s) = 0, \quad \varphi_2(s) = F(s, x, Y_s^{2,u,v}, Z_s^{2,u,v}, u_s, v_s) - F(s, X_s^{t,x;u,v}, Y_s^{2,u,v}, Z_s^{2,u,v}, u_s, v_s).$$

Obviously, the function g is Lipschitz with respect to (y, z) , and $|\varphi_2(s)| \leq C(1 + |x|^2)(|X_s^{t,x,u,v} - x| + |X_s^{t,x,u,v} - x|^3)$ for $s \in [t, t + \delta]$, $(t, x) \in [0, T] \times \mathbb{R}^n$, $u \in \mathcal{U}_{t,t+\delta}$, $v \in \mathcal{V}_{t,t+\delta}$. Thus, with the notation $\rho_0(r) = (1 + |x|^2)(r + r^3)$, $r \geq 0$, we have

$$\begin{aligned} & E \left[\int_t^{t+\delta} (|Y_s^{1,u,v} - Y_s^{2,u,v}|^2 + |Z_s^{1,u,v} - Z_s^{2,u,v}|^2) ds | \mathcal{F}_t \right] \\ & \leq CE \left[\int_t^{t+\delta} \rho_0^2(|X_s^{t,x,u,v} - x|) ds | \mathcal{F}_t \right] \\ & \leq C\delta E \left[\sup_{t \leq s \leq t+\delta} \rho_0^2(|X_s^{t,x,u,v} - x|) | \mathcal{F}_t \right] \\ & \leq C\delta^2. \end{aligned}$$

Therefore,

$$\begin{aligned} & |Y_t^{1,u,v} - Y_t^{2,u,v}| = |E[(Y_t^{1,u,v} - Y_t^{2,u,v}) | \mathcal{F}_t]| \\ & = \left| E \left[\int_t^{t+\delta} (F(s, X_s^{t,x,u,v}, Y_s^{1,u,v}, Z_s^{1,u,v}, u_s, v_s) - F(s, x, Y_s^{2,u,v}, Z_s^{2,u,v}, u_s, v_s)) ds | \mathcal{F}_t \right] \right| \\ & \leq CE \left[\int_t^{t+\delta} [\rho_0(|X_s^{t,x,u,v} - x|) + |Y_s^{1,u,v} - Y_s^{2,u,v}| + |Z_s^{1,u,v} - Z_s^{2,u,v}|] ds | \mathcal{F}_t \right] \\ & \leq CE \left[\int_t^{t+\delta} \rho_0(|X_s^{t,x,u,v} - x|) ds | \mathcal{F}_t \right] + C\delta^{\frac{1}{2}} \left\{ E \left[\int_t^{t+\delta} |Y_s^{1,u,v} - Y_s^{2,u,v}|^2 ds | \mathcal{F}_t \right] \right\}^{\frac{1}{2}} \\ & \quad + E \left[\int_t^{t+\delta} |Z_s^{1,u,v} - Z_s^{2,u,v}|^2 ds | \mathcal{F}_t \right]^{\frac{1}{2}} \Big\} \\ & \leq C\delta^{\frac{3}{2}}. \end{aligned}$$

Thus, the proof is complete. $\square$

LEMMA 4.5. Let $Y_0(\cdot)$ be the solution of the following ordinary differential equation:

$$(4.10) \quad \begin{cases} -\dot{Y}_0(s) &= F_0(s, x, Y_0(s), 0), \quad s \in [t, t + \delta], \\ Y_0(t + \delta) &= 0, \end{cases}$$

where the function F_0 is defined by

$$(4.11) \quad F_0(s, x, y, z) = \sup_{u \in U} \inf_{v \in V} F(s, x, y, z, u, v).$$

Then, P -a.s.,

$$(4.12) \quad \text{esssup}_{u \in \mathcal{U}_{t,t+\delta}} \text{essinf}_{v \in \mathcal{V}_{t,t+\delta}} Y_t^{2,u,v} = Y_0(t).$$

Proof. Obviously, $F_0(s, x, y, z)$ is Lipschitz in (y, z) , uniformly with respect to (s, x) . This guarantees existence and uniqueness for (4.10). We first introduce the function

$$(4.13) \quad F_1(s, x, y, z, u) = \inf_{v \in V} F(s, x, y, z, u, v), \quad (s, x, y, z, u) \in [0, T] \times \mathbb{R}^n \times \mathbb{R} \times \mathbb{R}^d \times U$$

and consider the BSDE

$$(4.14) \quad \begin{cases} -dY^{3,u}(s) &= F_1(s, x, Y^{3,u}(s), Z^{3,u}(s), u_s)ds - Z^{3,u}(s)dB_s, \\ Y^{3,u}(t+\delta) &= 0, \quad s \in [t, t+\delta], \end{cases}$$

for $u \in \mathcal{U}_{t,t+\delta}$. We notice that since $F_1(s, x, y, z, u_s)$ is Lipschitz in (y, z) , for every $u \in \mathcal{U}_{t,t+\delta}$, there exists a unique solution $(Y^{3,u}, Z^{3,u})$ to the BSDE (4.14). Moreover,

$$Y^{3,u}(t) = \text{essinf}_{v(\cdot) \in \mathcal{V}_{t,t+\delta}} Y_t^{2,u,v}, \quad P\text{-a.s.}, \quad \text{for any } u \in \mathcal{U}_{t,t+\delta}.$$

Indeed, from the definition of F_1 and Lemma 2.2 (comparison theorem) we have

$$Y^{3,u}(t) \leq \text{essinf}_{v(\cdot) \in \mathcal{V}_{t,t+\delta}} Y_t^{2,u,v}, \quad P\text{-a.s.}, \quad \text{for all } u \in \mathcal{U}_{t,t+\delta}.$$

On the other hand, there exists a measurable function $v^3 : [t, T] \times \mathbb{R}^n \times \mathbb{R} \times \mathbb{R}^d \times U \rightarrow V$ such that

$$F_1(s, x, y, z, u) = F(s, x, y, z, u, v^3(s, x, y, z, u)) \quad \text{for any } s, x, y, z, u.$$

We then put

$$\tilde{v}_s^3 := v^3(s, x, Y_s^{3,u}, Z_s^{3,u}, u_s), \quad s \in [t, t+\delta],$$

and we observe that $\tilde{v}^3 \in \mathcal{V}_{t,t+\delta}$, and

$$F_1(s, x, Y_s^{3,u}, Z_s^{3,u}, u_s) = F(s, x, Y_s^{3,u}, Z_s^{3,u}, u_s, \tilde{v}_s^3), \quad s \in [t, t+\delta].$$

Consequently, from the uniqueness of the solution of the BSDE it follows that $(Y^{3,u}, Z^{3,u}) = (Y^{2,u,\tilde{v}^3}, Z^{2,u,\tilde{v}^3})$ and, in particular, $Y_t^{3,u} = Y_t^{2,u,\tilde{v}^3}$, P -a.s., for any $u \in \mathcal{U}_{t,t+\delta}$. This proves that

$$Y^{3,u}(t) = \text{essinf}_{v \in \mathcal{V}_{t,t+\delta}} Y_t^{2,u,v}, \quad P\text{-a.s.}, \quad \text{for all } u \in \mathcal{U}_{t,t+\delta}.$$

Finally, since $F_0(s, x, y, z) = \sup_{u \in U} F_1(s, x, y, z, u)$, an argument similar to that developed above yields

$$Y_0(t) = \text{esssup}_{u \in \mathcal{U}_{t,t+\delta}} Y^{3,u}(t) (= \text{esssup}_{u \in \mathcal{U}_{t,t+\delta}} \text{essinf}_{v \in \mathcal{V}_{t,t+\delta}} Y_t^{2,u,v}), \quad P\text{-a.s.}$$

It uses the fact that (4.10) can be considered as a BSDE with the solution $(Y_s, Z_s) = (Y_0(s), 0)$. The proof is complete. $\square$

LEMMA 4.6. *For every $u \in \mathcal{U}_{t,t+\delta}$, $v \in \mathcal{V}_{t,t+\delta}$, we have*

$$(4.15) \quad E \left[\int_t^{t+\delta} |Y_s^{2,u,v}| ds | \mathcal{F}_t \right] + E \left[\int_t^{t+\delta} |Z_s^{2,u,v}| ds | \mathcal{F}_t \right] \leq C\delta^{\frac{3}{2}}, \quad P\text{-a.s.},$$

where the constant C is independent of the controls u, v .

Proof. Since $F(s, x, \cdot, \cdot, u, v)$ has a linear growth in (y, z) , uniformly in (u, v) , we get from Lemma 2.3 that, for some constant C independent of δ and the control processes u, v ,

$$|Y_s^{2,u,v}|^2 \leq C\delta, \quad E \left[\int_s^{t+\delta} |Z_r^{2,u,v}|^2 dr | \mathcal{F}_s \right] \leq C\delta, \quad s \in [t, t+\delta].$$

On the other hand, from (4.7),

$$\begin{aligned} |Y_s^{2,u,v}| &\leq E \left[\int_s^{t+\delta} |F(r, x, Y_r^{2,u,v}, Z_r^{2,u,v}, u_r, v_r)| dr | \mathcal{F}_s \right] \\ &\leq CE \left[\int_s^{t+\delta} (1 + |x|^2 + |Y_r^{2,u,v}| + |Z_r^{2,u,v}|) dr | \mathcal{F}_s \right] \\ &\leq C\delta + C\sqrt{\delta} \left(E \left[\int_s^{t+\delta} |Z_r^{2,u,v}|^2 dr | \mathcal{F}_s \right] \right)^{\frac{1}{2}} \leq C\delta, \quad P\text{-a.s.}, s \in [t, t + \delta], \end{aligned}$$

and, since

$$\int_t^{t+\delta} Z_s^{2,u,v} dB_s = \int_t^{t+\delta} F(s, x, Y_s^{2,u,v}, Z_s^{2,u,v}, u_s, v_s) ds - Y_t^{2,u,v},$$

we can get $E[\int_t^{t+\delta} |Z_s^{2,u,v}|^2 ds | \mathcal{F}_t] \leq C\delta^2$. Finally,

$$\begin{aligned} &E \left[\int_t^{t+\delta} |Y_s^{2,u,v}| ds | \mathcal{F}_t \right] + E \left[\int_t^{t+\delta} |Z_s^{2,u,v}| ds | \mathcal{F}_t \right] \\ &\leq C\delta^2 + \delta^{\frac{1}{2}} \left\{ E \left[\int_t^{t+\delta} |Z_s^{2,u,v}|^2 ds | \mathcal{F}_t \right] \right\}^{\frac{1}{2}} \leq C\delta^{\frac{3}{2}}. \end{aligned}$$

The proof is complete. $\square$

Now we are able to give the proof of Theorem 4.2.

Proof. (1) Obviously, $W(T, x) = \Phi(x)$, $x \in \mathbb{R}^n$. Let us show in a first step that W is a viscosity supersolution. For this we suppose that $\varphi \in C_{l,b}^3([0, T] \times \mathbb{R}^n)$ and $(t, x) \in [0, T] \times \mathbb{R}^n$ are such that $W - \varphi$ attains its minimum at (t, x) . Notice that we can replace the condition of a local minimum by that of a global one in the definition of the viscosity supersolution since W is continuous and of at most linear growth. Without loss of generality we may also suppose that $\varphi(t, x) = W(t, x)$. Then, due to the DPP (see Theorem 3.6),

$$\begin{aligned} \varphi(t, x) = W(t, x) &= \operatorname{essinf}_{\beta \in \mathcal{B}_{t,t+\delta}} \operatorname{esssup}_{u \in \mathcal{U}_{t,t+\delta}} G_{t,t+\delta}^{t,x;u,\beta(u)} [W(t + \delta, X_{t+\delta}^{t,x;u,\beta(u)})], \\ &0 \leq \delta \leq T - t, \end{aligned}$$

and from $W \geq \varphi$ and the monotonicity property of $G_{t,t+\delta}^{t,x;u,\beta(u)}[\cdot]$ (see Lemma 2.2) we obtain

$$\operatorname{essinf}_{\beta \in \mathcal{B}_{t,t+\delta}} \operatorname{esssup}_{u \in \mathcal{U}_{t,t+\delta}} \{G_{t,t+\delta}^{t,x;u,\beta(u)}[\varphi(t + \delta, X_{t+\delta}^{t,x;u,\beta(u)})] - \varphi(t, x)\} \leq 0, \quad P\text{-a.s.}$$

Thus, from Lemma 4.3,

$$\operatorname{essinf}_{\beta \in \mathcal{B}_{t,t+\delta}} \operatorname{esssup}_{u \in \mathcal{U}_{t,t+\delta}} Y_t^{1,u,\beta(u)} \leq 0, \quad P\text{-a.s.},$$

and further, from Lemma 4.4 we have

$$\operatorname{essinf}_{\beta \in \mathcal{B}_{t,t+\delta}} \operatorname{esssup}_{u \in \mathcal{U}_{t,t+\delta}} Y_t^{2,u,\beta(u)} \leq C\delta^{\frac{3}{2}}, \quad P\text{-a.s.}$$

Consequently, since $\operatorname{ess\,inf}_{v \in \mathcal{V}_{t,t+\delta}} Y_t^{2,u,v} \leq Y_t^{2,u,\beta(u)}$, $\beta \in \mathcal{B}_{t,t+\delta}$, we get

$$\operatorname{ess\,sup}_{u \in \mathcal{U}_{t,t+\delta}} \operatorname{ess\,inf}_{v \in \mathcal{V}_{t,t+\delta}} Y_t^{2,u,v} \leq \operatorname{ess\,inf}_{\beta \in \mathcal{B}_{t,t+\delta}} \operatorname{ess\,sup}_{u \in \mathcal{U}_{t,t+\delta}} Y_t^{2,u,\beta(u)} \leq C\delta^{\frac{3}{2}}, \quad P\text{-a.s.},$$

and Lemma 4.5 implies

$$Y_0(t) \leq C\delta^{\frac{3}{2}}, \quad P\text{-a.s.},$$

where Y_0 is the unique solution of (4.10). It then follows easily that

$$\sup_{u \in U} \inf_{v \in V} F(t, x, 0, 0, u, v) = F_0(t, x, 0, 0) \leq 0,$$

and from the definition of F we see that W is a viscosity supersolution of (4.1).

(2) The second step is devoted to the proof that W is a viscosity subsolution. For this we suppose that $\varphi \in C_{l,b}^3([0, T] \times \mathbb{R}^n)$ and $(t, x) \in [0, T] \times \mathbb{R}^n$ are such that $W - \varphi$ attains its maximum at (t, x) . Without loss of generality we suppose again that $\varphi(t, x) = W(t, x)$. We must prove that

$$\sup_{u \in U} \inf_{v \in V} F(t, x, 0, 0, u, v) = F_0(t, x, 0, 0) \geq 0.$$

Let us suppose that this is not true. Then there exists some $\theta > 0$ such that

$$(4.16) \quad F_0(t, x, 0, 0) = \sup_{u \in U} \inf_{v \in V} F(t, x, 0, 0, u, v) \leq -\theta < 0,$$

and we can find a measurable function $\psi : U \rightarrow V$ such that

$$F(t, x, 0, 0, u, \psi(u)) \leq -\frac{3}{4}\theta \text{ for all } u \in U.$$

Moreover, since $F(\cdot, x, 0, 0, \cdot, \cdot)$ is uniformly continuous on $[0, T] \times U \times V$, there exists some $T - t \geq R > 0$ such that

$$(4.17) \quad F(s, x, 0, 0, u, \psi(u)) \leq -\frac{1}{2}\theta \text{ for all } u \in U \text{ and } |s - t| \leq R.$$

On the other hand, due to the DPP (see Theorem 3.6), for every $\delta \in (0, R]$,

$$\varphi(t, x) = W(t, x) = \operatorname{ess\,inf}_{\beta \in \mathcal{B}_{t,t+\delta}} \operatorname{ess\,sup}_{u \in \mathcal{U}_{t,t+\delta}} G_{t,t+\delta}^{t,x;u,\beta(u)}[W(t + \delta, X_{t+\delta}^{t,x;u,\beta(u)})],$$

and from $W \leq \varphi$ and the monotonicity property of $G_{t,t+\delta}^{t,x;u,\beta(u)}[\cdot]$ (see Lemma 2.2) we obtain

$$\operatorname{ess\,inf}_{\beta \in \mathcal{B}_{t,t+\delta}} \operatorname{ess\,sup}_{u \in \mathcal{U}_{t,t+\delta}} \{G_{t,t+\delta}^{t,x;u,\beta(u)}[\varphi(t + \delta, X_{t+\delta}^{t,x;u,\beta(u)})] - \varphi(t, x)\} \geq 0, \quad P\text{-a.s.}$$

Thus, from Lemma 4.3,

$$\operatorname{ess\,inf}_{\beta \in \mathcal{B}_{t,t+\delta}} \operatorname{ess\,sup}_{u \in \mathcal{U}_{t,t+\delta}} Y_t^{1,u,\beta(u)} \geq 0, \quad P\text{-a.s.},$$

and, in particular,

$$\operatorname{ess\,sup}_{u \in \mathcal{U}_{t,t+\delta}} Y_t^{1,u,\psi(u)} \geq 0, \quad P\text{-a.s.}$$

Here, by putting $\psi_s(u)(\omega) = \psi(u_s(\omega))$, $(s, \omega) \in [t, T] \times \Omega$, we identify ψ as an element of $\mathcal{B}_{t,t+\delta}$. Given an arbitrarily $\varepsilon > 0$ we can choose $u^\varepsilon \in \mathcal{U}_{t,t+\delta}$ such that $Y_t^{1,u^\varepsilon,\psi(u^\varepsilon)} \geq -\varepsilon\delta$. From Lemma 4.4 we further have

$$(4.18) \quad Y_t^{2,u^\varepsilon,\psi(u^\varepsilon)} \geq -C\delta^{\frac{3}{2}} - \varepsilon\delta, \quad P\text{-a.s.}$$

Taking into account that

$$Y_t^{2,u^\varepsilon,\psi(u^\varepsilon)} = E \left[\int_t^{t+\delta} F(s, x, Y_s^{2,u^\varepsilon,\psi(u^\varepsilon)}, Z_s^{2,u^\varepsilon,\psi(u^\varepsilon)}, u_s^\varepsilon, \psi_s(u^\varepsilon)) ds | \mathcal{F}_t \right]$$

we get from the Lipschitz property of F in (y, z) , (4.17), and Lemma 4.6 that

$$(4.19) \quad \begin{aligned} Y_t^{2,u^\varepsilon,\psi(u^\varepsilon)} &\leq E \left[\int_t^{t+\delta} (C|Y_s^{2,u^\varepsilon,\psi(u^\varepsilon)}| + C|Z_s^{2,u^\varepsilon,\psi(u^\varepsilon)}| + F(s, x, 0, 0, u_s^\varepsilon, \psi_s(u^\varepsilon))) ds | \mathcal{F}_t \right] \\ &\leq C\delta^{\frac{3}{2}} - \frac{1}{2}\theta\delta, \quad P\text{-a.s.} \end{aligned}$$

From (4.18) and (4.19), $-C\delta^{\frac{1}{2}} - \varepsilon \leq C\delta^{\frac{1}{2}} - \frac{1}{2}\theta$, P -a.s. Letting $\delta \downarrow 0$, and then $\varepsilon \downarrow 0$, we deduce that $\theta \leq 0$, which induces a contradiction. Therefore,

$$F_0(t, x, 0, 0) = \sup_{u \in U} \inf_{v \in V} F(t, x, 0, 0, u, v) \geq 0,$$

and from the definition of F , we know that W is a viscosity subsolution of (4.1). Finally, the results from the first and the second steps prove that W is a viscosity solution of (4.1). $\square$

Remark 4.3. Similarly, we can prove that U is a viscosity solution of (4.2).

5. Viscosity solution of Isaacs' equation: Uniqueness theorem. The objective of this section is to study the uniqueness of the viscosity solution of Isaacs' equation (4.1):

$$(5.1) \quad \begin{cases} \frac{\partial}{\partial t} \omega(t, x) + H^-(t, x, \omega, D\omega, D^2\omega) = 0, & (t, x) \in [0, T] \times \mathbb{R}^n, \\ \omega(T, x) = \Phi(x), & x \in \mathbb{R}^n. \end{cases}$$

Recall that

$$H^-(t, x, y, p, X) = \sup_{u \in U} \inf_{v \in V} \left\{ \frac{1}{2} \text{tr}(\sigma \sigma^T(t, x, u, v) X) + p \cdot b(t, x, u, v) + f(t, x, y, p \cdot \sigma, u, v) \right\},$$

$t \in [0, T]$, $x \in \mathbb{R}^n$, $y \in \mathbb{R}$, $p \in \mathbb{R}^n$, and $X \in \mathbf{S}^n$. The functions b, σ, f , and Φ are still supposed to satisfy (H3.1) and (H3.2), respectively.

We will prove the uniqueness for (5.1) in the following space of continuous functions:

$$\Theta = \{ \varphi \in C([0, T] \times \mathbb{R}^n) : \exists \tilde{A} > 0 \text{ such that } \lim_{|x| \rightarrow \infty} \varphi(t, x) \exp\{-\tilde{A}[\log(|x|^2 + 1)]^{\frac{1}{2}}\} = 0, \text{ uniformly in } t \in [0, T] \}.$$

This space of continuous functions is endowed with a growth condition which is slightly weaker than the assumption of polynomial growth but more restrictive than that of exponential growth. This growth condition was introduced by Barles, Buckdahn, and

Pardoux [3] to prove the uniqueness of the viscosity solution of an integro-partial differential equation associated with a decoupled FBSDE with jumps. It was shown in [3] that this kind of growth condition is optimal for the uniqueness and can, in general, not be weakened. We adapt the ideas developed in [3] to Isaacs' equation (5.1) to prove the uniqueness of the viscosity solution in Θ . Since the proof of the uniqueness in Θ for (4.2) is the same, we will restrict ourselves to only that of (5.1). Before stating the main result of this section, let us begin with two auxiliary lemmas. Denoting by K a Lipschitz constant of $f(t, x, \cdot, \cdot, \cdot)$, that is, uniformly in (t, x) , we have the following.

LEMMA 5.1. *Let $u_1 \in \Theta$ be a viscosity subsolution and $u_2 \in \Theta$ be a viscosity supersolution of (5.1). Then the function $\omega := u_1 - u_2$ is a viscosity subsolution of the equation*

$$(5.2) \quad \begin{cases} \frac{\partial}{\partial t} \omega(t, x) + \sup_{u \in U, v \in V} \left\{ \frac{1}{2} \text{tr}(\sigma \sigma^T(t, x, u, v) D^2 \omega) + D\omega \cdot b(t, x, u, v) + K|\omega| \right. \\ \quad \left. + K|D\omega \cdot \sigma(t, x, u, v)| \right\} = 0, & (t, x) \in [0, T] \times \mathbb{R}^n, \\ \omega(T, x) = 0, & x \in \mathbb{R}^n. \end{cases}$$

The proof of this lemma follows directly that of Lemma 3.7 in [3], and it is even simpler because, contrary to Lemma 3.7 in [3], we do not have any integral part here in (5.1). In analogy to [3] we also have the following.

LEMMA 5.2. *For any $\tilde{A} > 0$, there exists $C_1 > 0$ such that the function*

$$\chi(t, x) = \exp[(C_1(T - t) + \tilde{A})\psi(x)],$$

with

$$\psi(x) = [\log((|x|^2 + 1)^{\frac{1}{2}}) + 1]^2, \quad x \in \mathbb{R}^n,$$

satisfies

$$(5.3) \quad \begin{aligned} & \frac{\partial}{\partial t} \chi(t, x) + \sup_{u \in U, v \in V} \left\{ \frac{1}{2} \text{tr}(\sigma \sigma^T(t, x, u, v) D^2 \chi) + D\chi \cdot b(t, x, u, v) + K\chi(t, x) \right. \\ & \quad \left. + K|D\chi(t, x) \cdot \sigma(t, x, u, v)| \right\} < 0 \quad \text{in } [t_1, T] \times \mathbb{R}^n, \quad \text{where } t_1 = T - \frac{\tilde{A}}{C_1}. \end{aligned}$$

Proof. By direct calculus we first deduce the following estimates for the first and second derivatives of ψ :

$$|D\psi(x)| \leq \frac{2[\psi(x)]^{\frac{1}{2}}}{(|x|^2 + 1)^{\frac{1}{2}}} \leq 4, \quad |D^2\psi(x)| \leq \frac{C(1 + [\psi(x)]^{\frac{1}{2}})}{|x|^2 + 1}, \quad x \in \mathbb{R}^n.$$

These estimates imply that, if $t \in [t_1, T]$,

$$\begin{aligned} |D\chi(t, x)| & \leq (C_1(T - t) + \tilde{A})\chi(t, x)|D\psi(x)| \\ & \leq C\chi(t, x) \frac{[\psi(x)]^{\frac{1}{2}}}{(|x|^2 + 1)^{\frac{1}{2}}}, \end{aligned}$$

and, similarly,

$$|D^2\chi(t, x)| \leq C\chi(t, x) \frac{\psi(x)}{|x|^2 + 1}.$$

We should notice that the above estimates do not depend on C_1 because of the definition of t_1 . By virtue with the above estimates we have

$$\begin{aligned} & \frac{\partial}{\partial t} \chi(t, x) + \sup_{u \in U, v \in V} \left\{ \frac{1}{2} \text{tr}(\sigma \sigma^T(t, x, u, v) D^2 \chi) + D\chi \cdot b(t, x, u, v) + K\chi(t, x) \right. \\ & \quad \left. + K|D\chi(t, x) \cdot \sigma(t, x, u, v)| \right\} \\ & \leq -\chi(t, x) \{C_1 \psi(x) - C\psi(x) - C[\psi(x)]^{\frac{1}{2}} - K\} \\ & < -\chi(t, x) \{C_1 - [2C + K]\} \psi(x) < 0, \text{ if } C_1 > 2C + K \text{ large enough.} \quad \square \end{aligned}$$

Now we can prove the uniqueness theorem.

THEOREM 5.3. *We assume that (H3.1) and (H3.2) hold. Let u_1 (resp., u_2) $\in \Theta$ be a viscosity subsolution (resp., supersolution) of (5.1). Then we have*

$$(5.4) \quad u_1(t, x) \leq u_2(t, x) \quad \text{for all } (t, x) \in [0, T] \times \mathbb{R}^n.$$

Proof. Let us put $\omega := u_1 - u_2$. Then we have, for some $\tilde{A} > 0$,

$$\lim_{|x| \rightarrow \infty} \omega(t, x) e^{-\tilde{A}[\log((|x|^2+1)^{\frac{1}{2}})]^2} = 0,$$

uniformly with respect to $t \in [0, T]$. This implies, in particular, that, for any $\alpha > 0$, $\omega(t, x) - \alpha\chi(t, x)$ is bounded from above in $[t_1, T] \times \mathbb{R}^n$ and that

$$M := \max_{[t_1, T] \times \mathbb{R}^n} (\omega - \alpha\chi)(t, x) e^{-K(T-t)}$$

is achieved at some point $(t_0, x_0) \in [t_1, T] \times \mathbb{R}^n$ (depending on α). We now have to distinguish between two cases.

For the first case we suppose that: $\omega(t_0, x_0) \leq 0$, for any $\alpha > 0$.

Then, obviously $M \leq 0$ and $u_1(t, x) - u_2(t, x) \leq \alpha\chi(t, x)$ in $[t_1, T] \times \mathbb{R}^n$. Consequently, letting α tend to zero we obtain

$$u_1(t, x) \leq u_2(t, x) \quad \text{for all } (t, x) \in [t_1, T] \times \mathbb{R}^n.$$

For the second case we assume that there exists some $\alpha > 0$ such that $\omega(t_0, x_0) > 0$.

We notice that $\omega(t, x) - \alpha\chi(t, x) \leq (\omega(t_0, x_0) - \alpha\chi(t_0, x_0))e^{-K(t-t_0)}$ in $[t_1, T] \times \mathbb{R}^n$. Then, putting

$$\varphi(t, x) = \alpha\chi(t, x) + (\omega - \alpha\chi)(t_0, x_0)e^{-K(t-t_0)}$$

we get $\omega - \varphi \leq 0 = (\omega - \varphi)(t_0, x_0)$ in $[t_1, T] \times \mathbb{R}^n$. Consequently, since ω is a viscosity subsolution of (5.2), from Lemma 5.1 we have

$$\begin{aligned} & \frac{\partial}{\partial t} \varphi(t_0, x_0) + \sup_{u \in U, v \in V} \left\{ \frac{1}{2} \text{tr}(\sigma \sigma^T(t_0, x_0, u, v) D^2 \varphi(t_0, x_0)) + D\varphi(t_0, x_0) \cdot b(t_0, x_0, u, v) \right. \\ & \quad \left. + K|\varphi(t_0, x_0)| + K|D\varphi(t_0, x_0) \cdot \sigma(t_0, x_0, u, v)| \right\} \geq 0. \end{aligned}$$

Moreover, due to our assumption that $\omega(t_0, x_0) > 0$ and since $\omega(t_0, x_0) = \varphi(t_0, x_0)$, we can replace $K|\varphi(t_0, x_0)|$ by $K\varphi(t_0, x_0)$ in the above formula. Then, from the definition of φ and Lemma 5.2,

$$0 \leq \alpha \left\{ \frac{\partial \chi}{\partial t}(t_0, x_0) + \sup_{u \in U, v \in V} \left\{ \frac{1}{2} \text{tr}(\sigma \sigma^T(t_0, x_0, u, v) D^2 \chi(t_0, x_0)) + D\chi(t_0, x_0) \cdot b(t_0, x_0, u, v) \right. \right. \\ \left. \left. + K\chi(t_0, x_0) + K|D\chi(t_0, x_0) \cdot \sigma(t_0, x_0, u, v)| \right\} \right\} < 0,$$

which is a contradiction. Finally, by applying successively the same argument on the interval $[t_2, t_1]$, with $t_2 = (t_1 - \frac{\tilde{A}}{C_1})^+$, and then, if $t_2 > 0$, on $[t_3, t_2]$, with $t_3 = (t_2 - \frac{\tilde{A}}{C_1})^+$, etc., we get

$$u_1(t, x) \leq u_2(t, x), \quad (t, x) \in [0, T] \times \mathbb{R}^n.$$

Thus, the proof is complete. $\square$

Remark 5.1. Obviously, since the lower value function $W(t, x)$ is of at most linear growth, it belongs to Θ , and so $W(t, x)$ is the unique viscosity solution in Θ of (5.1). Similarly we get that the upper value function $U(t, x)$ is the unique viscosity solution in Θ of (4.2). On the other hand, since $H^- \leq H^+$, any viscosity solution of (4.2) is a supersolution of (5.1). Then, again from Theorem 5.3, it follows that $W \leq U$. This justifies calling W the lower value function and U the upper value function.

Remark 5.2. If the Isaacs' condition holds, that is, if for all $(t, x, y, p, X) \in [0, T] \times \mathbb{R}^n \times \mathbb{R} \times \mathbb{R}^n \times \mathbf{S}^n$

$$H^-(t, x, y, p, X) = H^+(t, x, y, p, X),$$

then (5.1) and (4.2) coincide, and from the uniqueness in Θ of viscosity solution it follows that the lower value function $W(t, x)$ equals the upper value function $U(t, x)$ which means that the associated stochastic differential game has a value.

Remark 5.3. Let us assume that the coefficient of BSDE (3.5) $f(t, x, y, z, u, v) \equiv f(t, x, u, v)$ is independent of (y, z) and denote by $\tilde{W}(t, x)$ (resp., $\tilde{U}(t, x)$) the lower value function (resp., the upper value function) defined by Fleming and Souganidis [14]; see Remark 3.4. It is shown in [14] that $\tilde{W}(t, x)$ is a viscosity solution in Θ of (5.1) and $\tilde{U}(t, x)$ a viscosity solution in Θ of (4.2). Then, due to Theorem 5.3, $W(t, x) = \tilde{W}(t, x)$ and $U(t, x) = \tilde{U}(t, x)$, $(t, x) \in [0, T] \times \mathbb{R}^n$. Moreover, if the Isaacs' condition holds, then $W(t, x) = \tilde{W}(t, x) = \tilde{U}(t, x) = U(t, x)$.

Appendix A. Forward-backward SDEs (FBSDEs).

In this section we give an overview over basic results which are necessary for us on BSDEs associated with forward SDEs. We consider measurable functions $b : [0, T] \times \Omega \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ and $\sigma : [0, T] \times \Omega \times \mathbb{R}^n \rightarrow \mathbb{R}^{n \times d}$, which are supposed to satisfy the following conditions:

(H6.1)

- (i) $b(\cdot, 0)$ and $\sigma(\cdot, 0)$ are $\mathcal{F}_t$ -adapted processes, and there exists some constant $C > 0$ such that $|b(t, x)| + |\sigma(t, x)| \leq C(1 + |x|)$, a.s., for all $0 \leq t \leq T$, $x \in \mathbb{R}^n$;
- (ii) b and σ are Lipschitz in x ; i.e., there is some constant $C > 0$ such that

$$|b(t, x) - b(t, x')| + |\sigma(t, x) - \sigma(t, x')| \leq C|x - x'|, \text{ a.s.,} \\ \text{for all } 0 \leq t \leq T, \ x, \ x' \in \mathbb{R}^n.$$

We now consider the following SDE parameterized by the initial condition $(t, \zeta) \in [0, T] \times L^2(\Omega, \mathcal{F}_t, P; \mathbb{R}^n)$:

$$(A.1) \quad \begin{cases} dX_s^{t, \zeta} &= b(s, X_s^{t, \zeta})ds + \sigma(s, X_s^{t, \zeta})dB_s, \quad s \in [t, T], \\ X_t^{t, \zeta} &= \zeta. \end{cases}$$

Under the assumption (H6.1), SDE (A.1) has a unique strong solution, and, for any $p \geq 2$, there exists $C_p \in \mathbb{R}$ such that, for any $t \in [0, T]$ and $\zeta, \zeta' \in L^p(\Omega, \mathcal{F}_t, P; \mathbb{R}^n)$,

$$(A.2) \quad \begin{aligned} E \left[\sup_{t \leq s \leq T} |X_s^{t, \zeta} - X_s^{t, \zeta'}|^p | \mathcal{F}_t \right] &\leq C_p |\zeta - \zeta'|^p, \quad a.s., \\ E \left[\sup_{t \leq s \leq T} |X_s^{t, \zeta}|^p | \mathcal{F}_t \right] &\leq C_p (1 + |\zeta|^p), \quad a.s. \end{aligned}$$

These well-known standard estimates can be consulted, for instance, in Ikeda and Watanabe [18, pp. 166–168] and also in Karatzas and Shreve [19, pp. 289–290].

We emphasize that the constant C_p in (A.2) depends only on the Lipschitz and the growth constants of b and σ . Let now be given two real-valued functions $f(t, x, y, z)$ and $\Phi(x)$ which shall satisfy the following conditions:

(H6.2)

- (i) $\Phi : \Omega \times \mathbb{R}^n \rightarrow \mathbb{R}$ is an $\mathcal{F}_T \otimes \mathcal{B}(\mathbb{R}^n)$ -measurable random variable and $f : [0, T] \times \Omega \times \mathbb{R}^n \times \mathbb{R} \times \mathbb{R}^d \rightarrow \mathbb{R}$ is a measurable process such that $f(\cdot, x, y, z)$ is $\mathcal{F}_t$ -adapted, for all $(x, y, z) \in \mathbb{R}^n \times \mathbb{R} \times \mathbb{R}^d$;
- (ii) There exists a constant $C > 0$ such that $|f(t, x, y, z) - f(t, x', y', z')| + |\Phi(x) - \Phi(x')| \leq C(|x - x'| + |y - y'| + |z - z'|)$, *a.s.*, for all $0 \leq t \leq T$, $x, x' \in \mathbb{R}^n$, $y, y' \in \mathbb{R}$ and $z, z' \in \mathbb{R}^d$;
- (iii) f and Φ satisfy a linear growth condition; i.e., there exists some $C > 0$ such that, $dt \times dP$ -a.e., for all $x \in \mathbb{R}^n$, $|f(t, x, 0, 0)| + |\Phi(x)| \leq C(1 + |x|)$.

With the help of the above assumptions we can verify that the coefficient $f(s, X_s^{t, \zeta}, y, z)$ satisfies the hypotheses (A.1) and (A.2) and $\xi = \Phi(X_T^{t, \zeta}) \in L^2(\Omega, \mathcal{F}_T, P; \mathbb{R})$. Therefore, the following BSDE possesses a unique solution:

$$(A.3) \quad \begin{cases} -dY_s^{t, \zeta} &= f(s, X_s^{t, \zeta}, Y_s^{t, \zeta}, Z_s^{t, \zeta})ds - Z_s^{t, \zeta}dB_s, \quad s \in [t, T], \\ Y_T^{t, \zeta} &= \Phi(X_T^{t, \zeta}). \end{cases}$$

PROPOSITION A.1. *We suppose that the hypotheses (H6.1) and (H6.2) hold. Then, for any $0 \leq t \leq T$ and the associated initial conditions $\zeta, \zeta' \in L^2(\Omega, \mathcal{F}_t, P; \mathbb{R}^n)$, we have the following estimates:*

- (i) $E \left[\sup_{t \leq s \leq T} |Y_s^{t, \zeta}|^2 + \int_t^T |Z_s^{t, \zeta}|^2 ds | \mathcal{F}_t \right] \leq C(1 + |\zeta|^2), \quad a.s.;$
- (ii) $E \left[\sup_{t \leq s \leq T} |Y_s^{t, \zeta} - Y_s^{t, \zeta'}|^2 + \int_t^T |Z_s^{t, \zeta} - Z_s^{t, \zeta'}|^2 ds | \mathcal{F}_t \right] \leq C|\zeta - \zeta'|^2, \quad a.s.$

In particular,

$$(A.4) \quad \text{(iii)} \quad |Y_t^{t, \zeta}| \leq C(1 + |\zeta|), \quad a.s.; \quad \text{(iv)} \quad |Y_t^{t, \zeta} - Y_t^{t, \zeta'}| \leq C|\zeta - \zeta'|, \quad a.s.,$$

where the constant $C > 0$ depends only on the Lipschitz and the growth constants of b , σ , f , and Φ .

For the proof the reader is referred to Proposition 4.1 of Peng [26]; a similar result can be found in El Karoui, Peng, and Quenez [12, Proposition 4.1].

Let us now introduce the random field:

$$(A.5) \quad u(t, x) = Y_s^{t,x}|_{s=t}, \quad (t, x) \in [0, T] \times \mathbb{R}^n,$$

where $Y^{t,x}$ is the solution of BSDE (A.3) with $x \in \mathbb{R}^n$ at the place of $\zeta \in L^2(\Omega, \mathcal{F}_t, P; \mathbb{R}^n)$.

As a consequence of Proposition A.1 we have that, for all $t \in [0, T]$, P -a.s.,

$$(A.6) \quad \begin{aligned} (i) \quad & |u(t, x) - u(t, y)| \leq C|x - y| \text{ for all } x, y \in \mathbb{R}^n; \\ (ii) \quad & |u(t, x)| \leq C(1 + |x|) \text{ for all } x \in \mathbb{R}^n. \end{aligned}$$

Remark A.1. In the general situation u is an adapted random function; that is, for any $x \in \mathbb{R}^n$, $u(\cdot, x)$ is an $\mathcal{F}_t$ -adapted real-valued process. Indeed, recall that b , σ , and f all are $\mathbb{F}$ -adapted random functions while Φ is $\mathcal{F}_T$ -measurable. On the other hand, it is well known that, under the additional assumption that the functions

$$(H6.3) \quad b, \sigma, f, \text{ and } \Phi \text{ are deterministic,}$$

u is also a deterministic function of (t, x) .

The random field u and $Y^{t,\zeta}$, $(t, \zeta) \in [0, T] \times L^2(\Omega, \mathcal{F}_t, P; \mathbb{R}^n)$, are related by the following theorem.

THEOREM A.2. *Under the assumptions (H6.1) and (H6.2), for any $t \in [0, T]$ and $\zeta \in L^2(\Omega, \mathcal{F}_t, P; \mathbb{R}^n)$, we have*

$$(A.7) \quad u(t, \zeta) = Y_t^{t,\zeta}, \quad P\text{-a.s.}$$

The proof of Theorem A.2 can be found in Peng [26]; we give it for the reader's convenience. It makes use of the following definition.

DEFINITION A.3. *For any $t \in [0, T]$, a sequence $\{A_i\}_{i=1}^N \subset \mathcal{F}_t$ (with $1 \leq N \leq \infty$) is called a partition of $(\Omega, \mathcal{F}_t)$ if $\cup_{i=1}^N A_i = \Omega$ and $A_i \cap A_j = \emptyset$, whenever $i \neq j$.*

Proof of Theorem A.2. We first consider the case where ζ is a simple random variable of the form

$$(A.8) \quad \zeta = \sum_{i=1}^N x_i \mathbf{1}_{A_i},$$

where $\{A_i\}_{i=1}^N$ is a finite partition of $(\Omega, \mathcal{F}_t)$ and $x_i \in \mathbb{R}^n$, for $1 \leq i \leq N$.

For each i , we put $(X_s^i, Y_s^i, Z_s^i) \equiv (X_s^{t,x_i}, Y_s^{t,x_i}, Z_s^{t,x_i})$. Then X^i is the solution of the SDE

$$X_s^i = x_i + \int_t^s b(r, X_r^i) dr + \int_t^s \sigma(r, X_r^i) dB_r, \quad s \in [t, T],$$

and (Y^i, Z^i) is the solution of the associated BSDE

$$Y_s^i = \Phi(X_T^i) + \int_s^T f(r, X_r^i, Y_r^i, Z_r^i) dr - \int_s^T Z_r^i dB_r, \quad s \in [t, T].$$

The above two equations are multiplied by $\mathbf{1}_{A_i}$ and summed up with respect to i . Thus, taking into account that $\sum_i \varphi(x_i) \mathbf{1}_{A_i} = \varphi(\sum_i x_i \mathbf{1}_{A_i})$, we get

$$\sum_{i=1}^N \mathbf{1}_{A_i} X_s^i = \sum_{i=1}^N x_i \mathbf{1}_{A_i} + \int_t^s b \left(r, \sum_{i=1}^N \mathbf{1}_{A_i} X_r^i \right) dr + \int_t^s \sigma \left(r, \sum_{i=1}^N \mathbf{1}_{A_i} X_r^i \right) dB_r$$

and

$$\begin{aligned} \sum_{i=1}^N \mathbf{1}_{A_i} Y_s^i &= \Phi \left(\sum_{i=1}^N \mathbf{1}_{A_i} X_T^i \right) + \int_s^T f \left(r, \sum_{i=1}^N \mathbf{1}_{A_i} X_r^i, \sum_{i=1}^N \mathbf{1}_{A_i} Y_r^i, \sum_{i=1}^N \mathbf{1}_{A_i} Z_r^i \right) dr \\ &\quad - \int_s^T \sum_{i=1}^N \mathbf{1}_{A_i} Z_r^i dB_r. \end{aligned}$$

Then the strong uniqueness property of the solution of the SDE and the BSDE yields

$$X_s^{t,\zeta} = \sum_{i=1}^N X_s^i \mathbf{1}_{A_i}, \quad (Y_s^{t,\zeta}, Z_s^{t,\zeta}) = \left(\sum_{i=1}^N \mathbf{1}_{A_i} Y_s^i, \sum_{i=1}^N \mathbf{1}_{A_i} Z_s^i \right), \quad s \in [t, T].$$

Finally, from $u(t, x_i) = Y_t^i$, $1 \leq i \leq N$, we deduce that

$$Y_t^{t,\zeta} = \sum_{i=1}^N Y_t^i \mathbf{1}_{A_i} = \sum_{i=1}^N u(t, x_i) \mathbf{1}_{A_i} = u \left(t, \sum_{i=1}^N x_i \mathbf{1}_{A_i} \right) = u(t, \zeta).$$

Therefore, for simple random variables, we have the desired result.

Given a general $\zeta \in L^2(\Omega, \mathcal{F}_t, P; \mathbb{R}^n)$ we can choose a sequence of simple random variables $\{\zeta_i\}$ which converges to ζ in $L^2(\Omega, \mathcal{F}_t, P; \mathbb{R}^n)$. Consequently, from the estimates (A.4) and (A.6) and the first step of the proof, we have

$$\begin{aligned} E|Y_t^{t,\zeta_i} - Y_t^{t,\zeta}|^2 &\leq CE|\zeta_i - \zeta|^2 \rightarrow 0, \quad i \rightarrow \infty, \\ E|u(t, \zeta_i) - u(t, \zeta)|^2 &\leq CE|\zeta_i - \zeta|^2 \rightarrow 0, \quad i \rightarrow \infty, \\ \text{and} \quad Y_t^{t,\zeta_i} &= u(t, \zeta_i), \quad i \geq 1. \end{aligned}$$

Then the proof is complete. $\square$

Remark A.2. Under (H6.1), (H6.2), and (H6.3) we know $u(t, x)$ is $\frac{1}{2}$ -Hölder continuous in t : There exists a constant C such that, for every $x \in \mathbb{R}^n$, $t, t' \in [0, T]$,

$$|u(t, x) - u(t', x)| \leq C(1 + |x|)|t - t'|^{\frac{1}{2}}.$$

This inequality can be proved with the help of Theorem A.2. Since, on the other hand, a similar result but in a more general setting is proved (see Theorem 3.10), we do not give the proof here.

Although it is not used in our paper, let us also mention that, in the case of random coefficients b, σ, f , and Φ the following holds true.

Remark A.3. Let us suppose in addition to the assumptions (H6.1) and (H6.2) that $\sigma(\omega, t, \cdot)$ and $b(\omega, t, \cdot)$ are continuously differentiable with a Lipschitz derivative such that, for some constant C ,

$$\begin{aligned} |D_x \sigma(\omega, t, x)| + |D_x b(\omega, t, x)| &\leq C, \quad \text{dtd}P\text{-a.e., for all } x \in \mathbb{R}^n; \\ D_x \sigma(\omega, t, \cdot), D_x b(\omega, t, \cdot) &\text{ are Lipschitz, uniformly in } (\omega, t). \end{aligned}$$

Then the random field $u(\omega, t, x) : \Omega \times [0, T] \times \mathbb{R}^n \rightarrow \mathbb{R}$ possesses a continuous version.

The proof of the above property uses a standard argument for BSDEs based on the fact that $u(s, x)$ can be written as $Y_s^{t, \bar{X}_s^{t,x}}$ for $0 < s - t$ small enough, where $\bar{X}_s^{t,x}$ denotes the local inversion of the stochastic flow generated by (A.1): $\bar{X}_s^{t,x} = (X_s^{t, \cdot})^{-1}(x)$.

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